Statistical Rethinking

2023

18. Missing Data

Missing Data, Found

Observed data is special case: We trick ourselves into believing there is no error

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Observed data is special case: We trick ourselves into believing there is no error

Most data are missing most of the time

"Missing" data: Some cases unobserved

Not totally "missing": We know
(1) constraints
(2) relationships to other variables

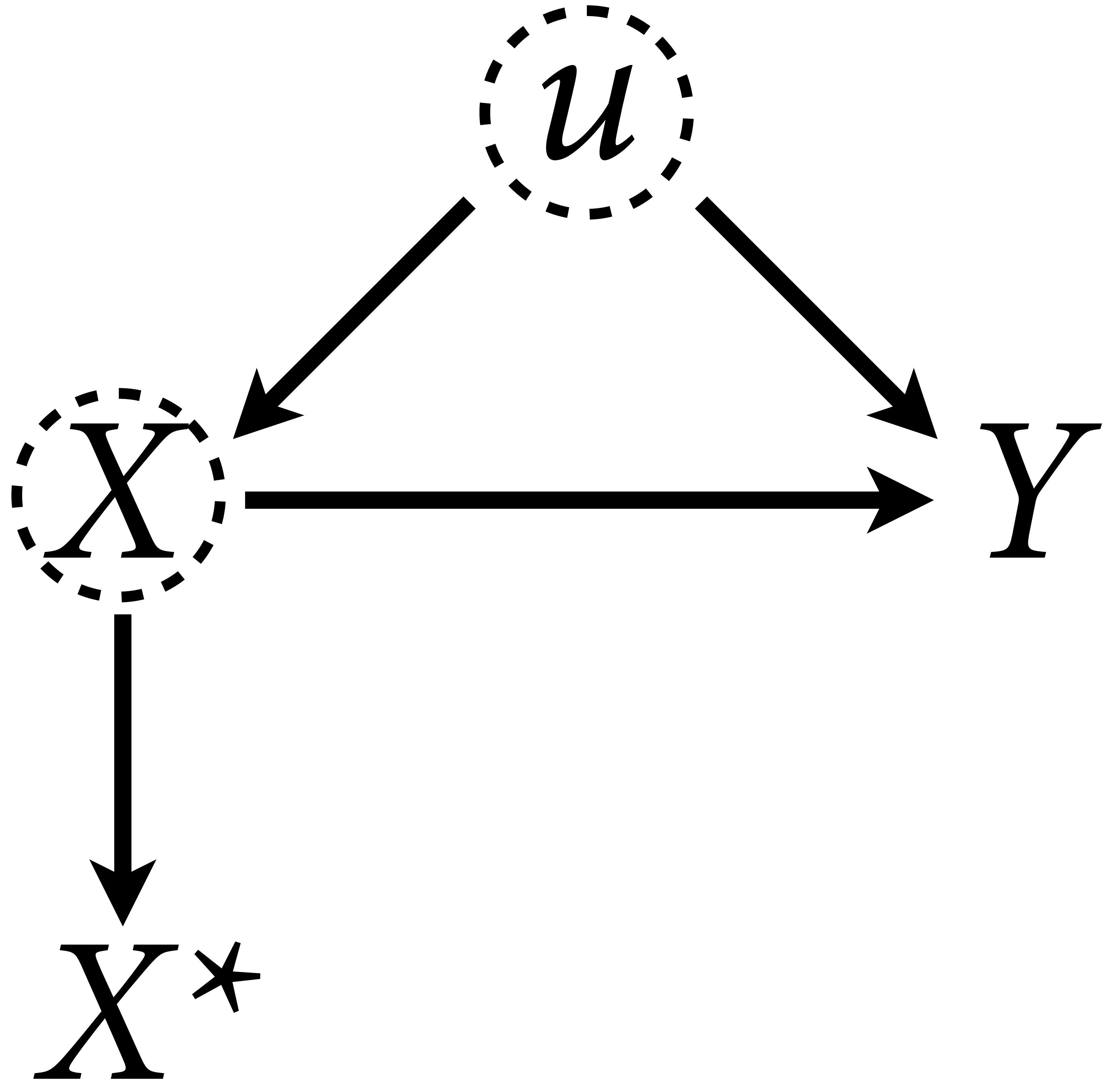
Missing Data is Workflow

What to do with missing data?

Dropping cases with missing values sometimes justifiable

Right thing to do depends upon causal assumptions

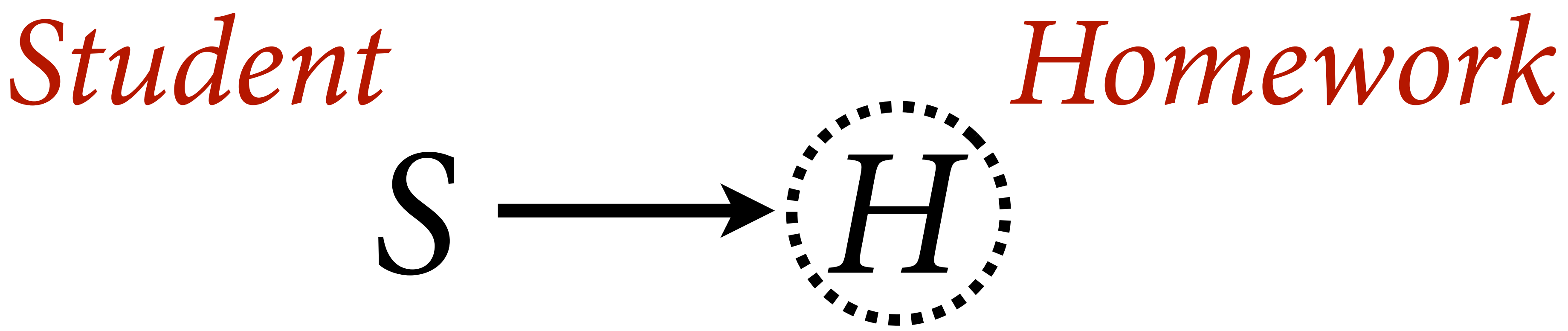
Imputation often beneficial/necessary

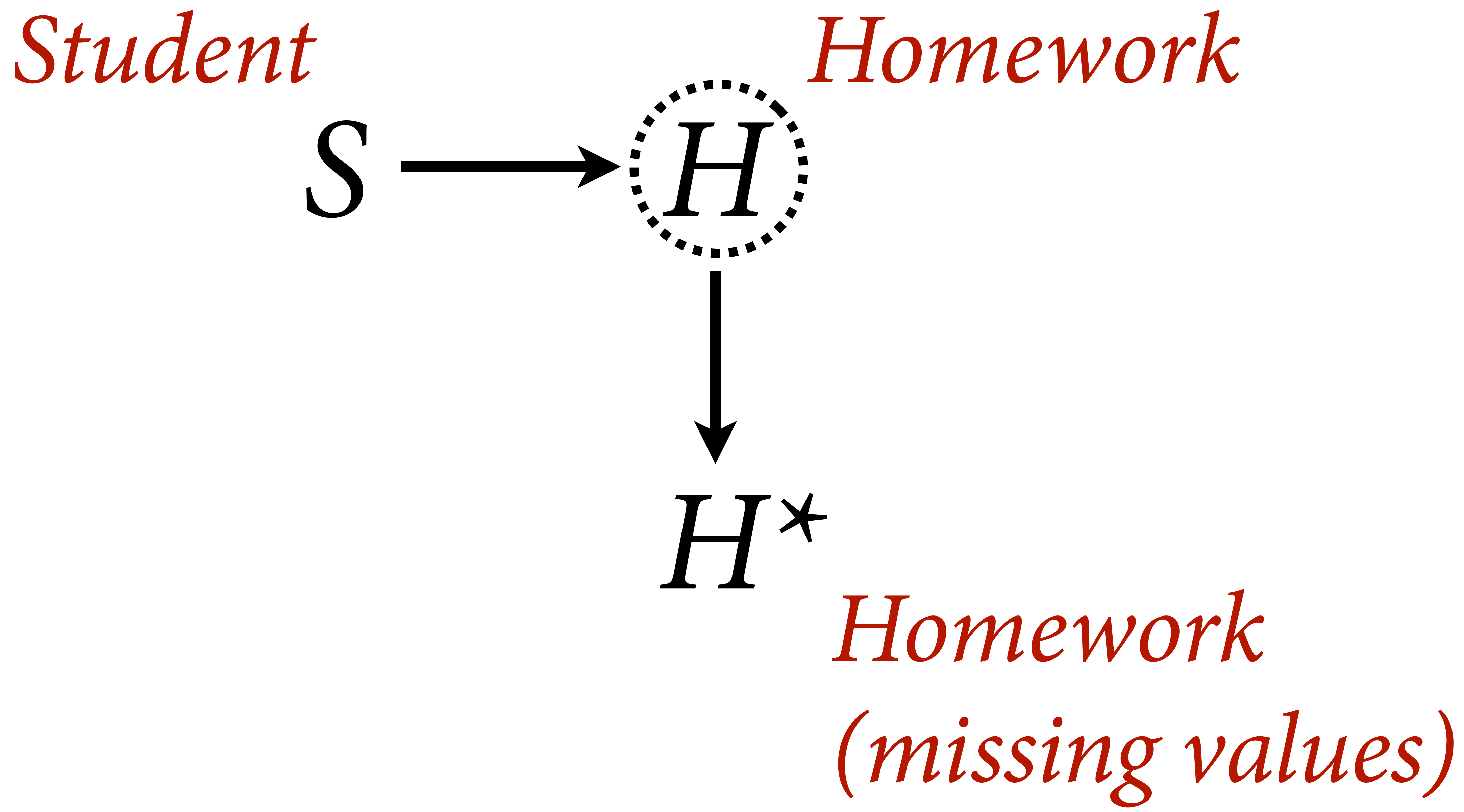


Student

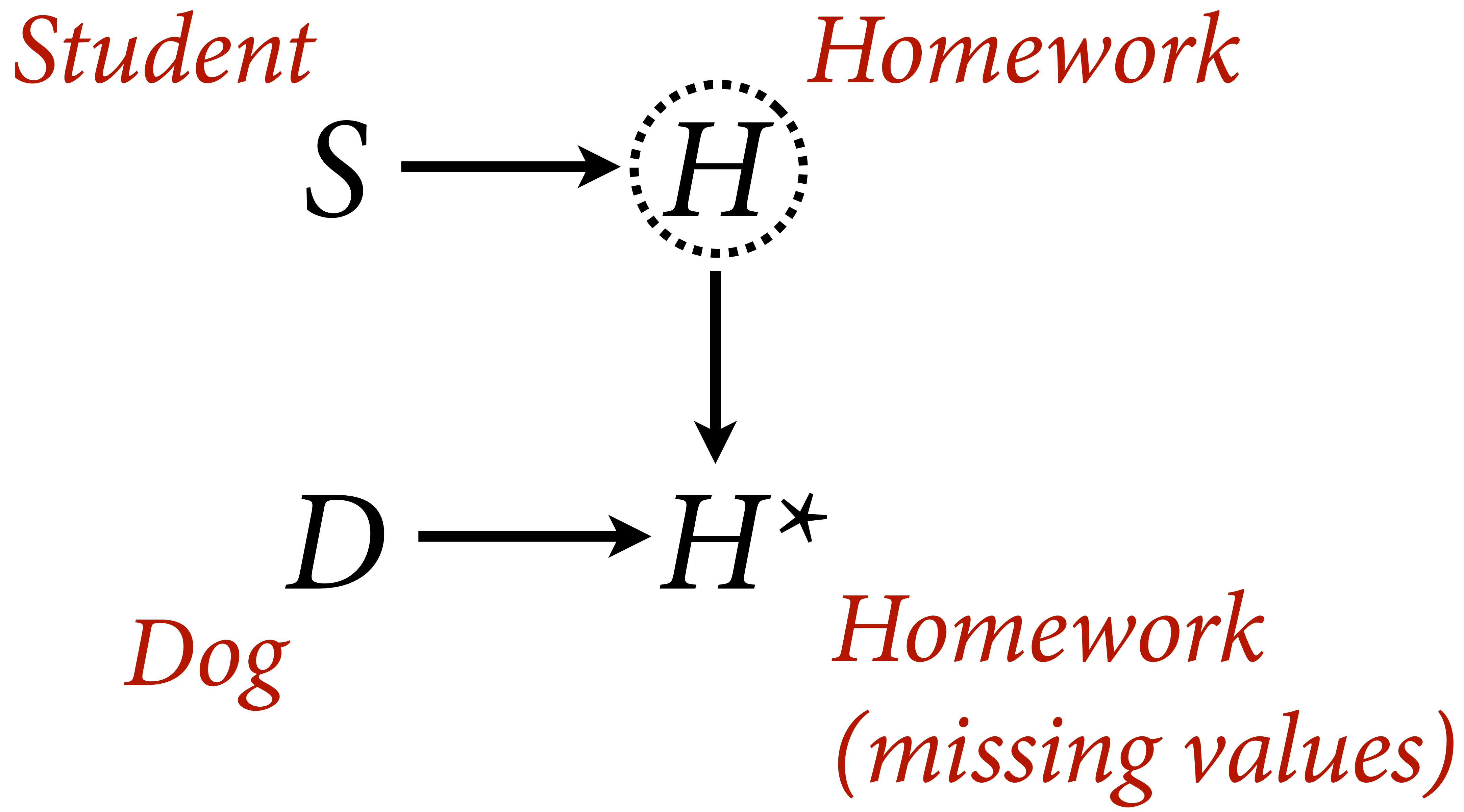
$$S \longrightarrow H$$

Homework

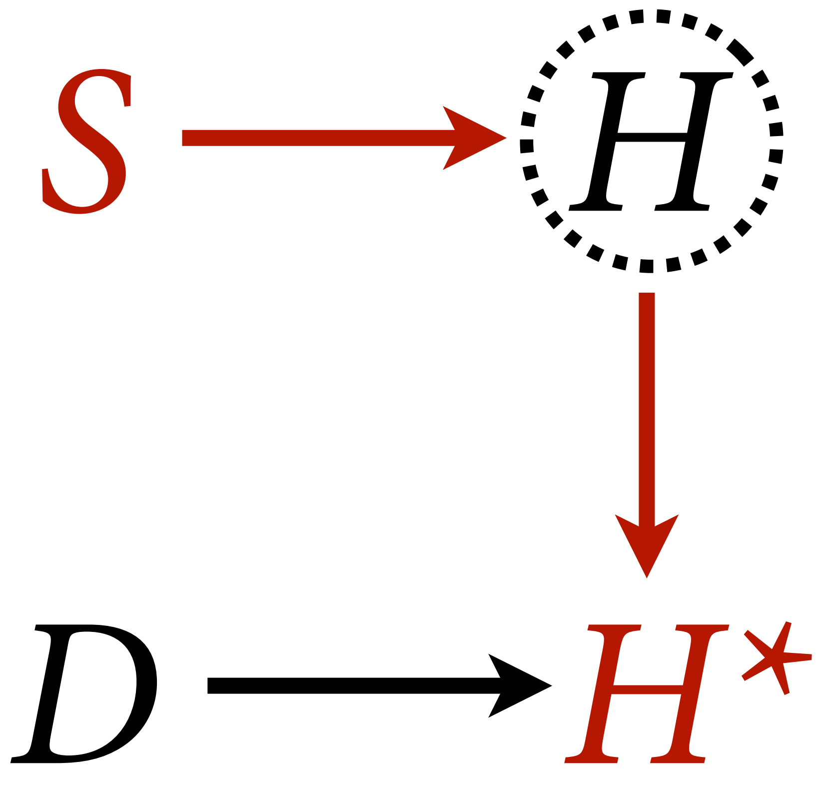

Student
Homework
S
H



Student
S
H
Homework
H*
Homework
(missing values)


Student
S
H
Homework
D
H*
Dog
Homework
(missing values)

No biasing paths connecting H to S



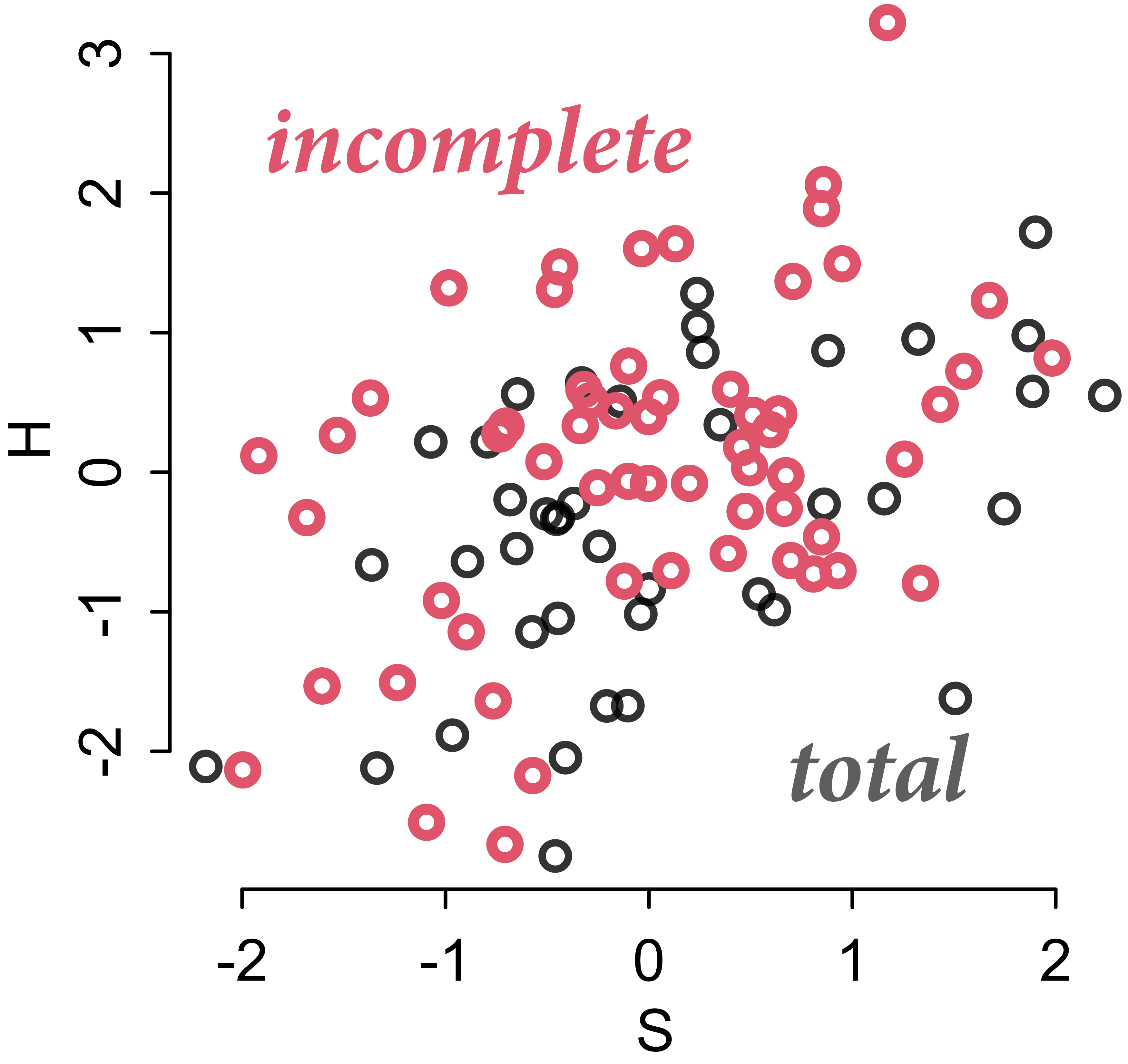
"Dog eats random homework"

"Missing completely at random"

Dog usually benign

```
# Dog eats random homework
# aka missing completely at random
N <- 100
S <- rnorm(N)
H <- rnorm(N,0.5*S)
# dog eats 50% of homework at random
D <- rbern(N,0.5)
Hstar <- H
Hstar[D==1] <- NA

plot( S , H , col=grau(0.8) , lwd=2 )
points( S , Hstar , col=2 , lwd=3 )
```

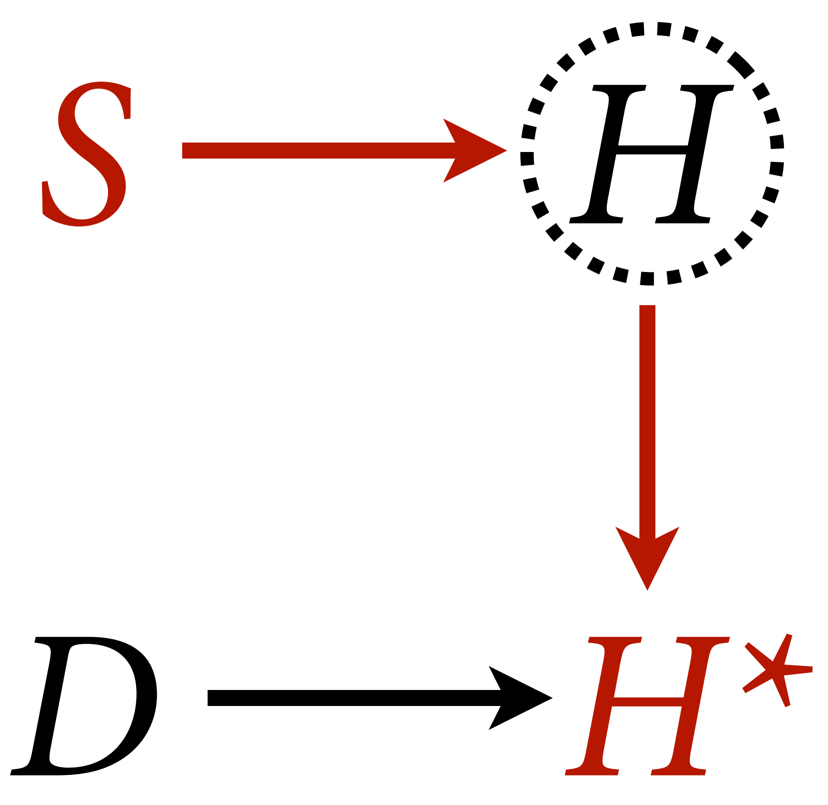
$$S \rightarrow H \rightarrow H^* \leftarrow D$$

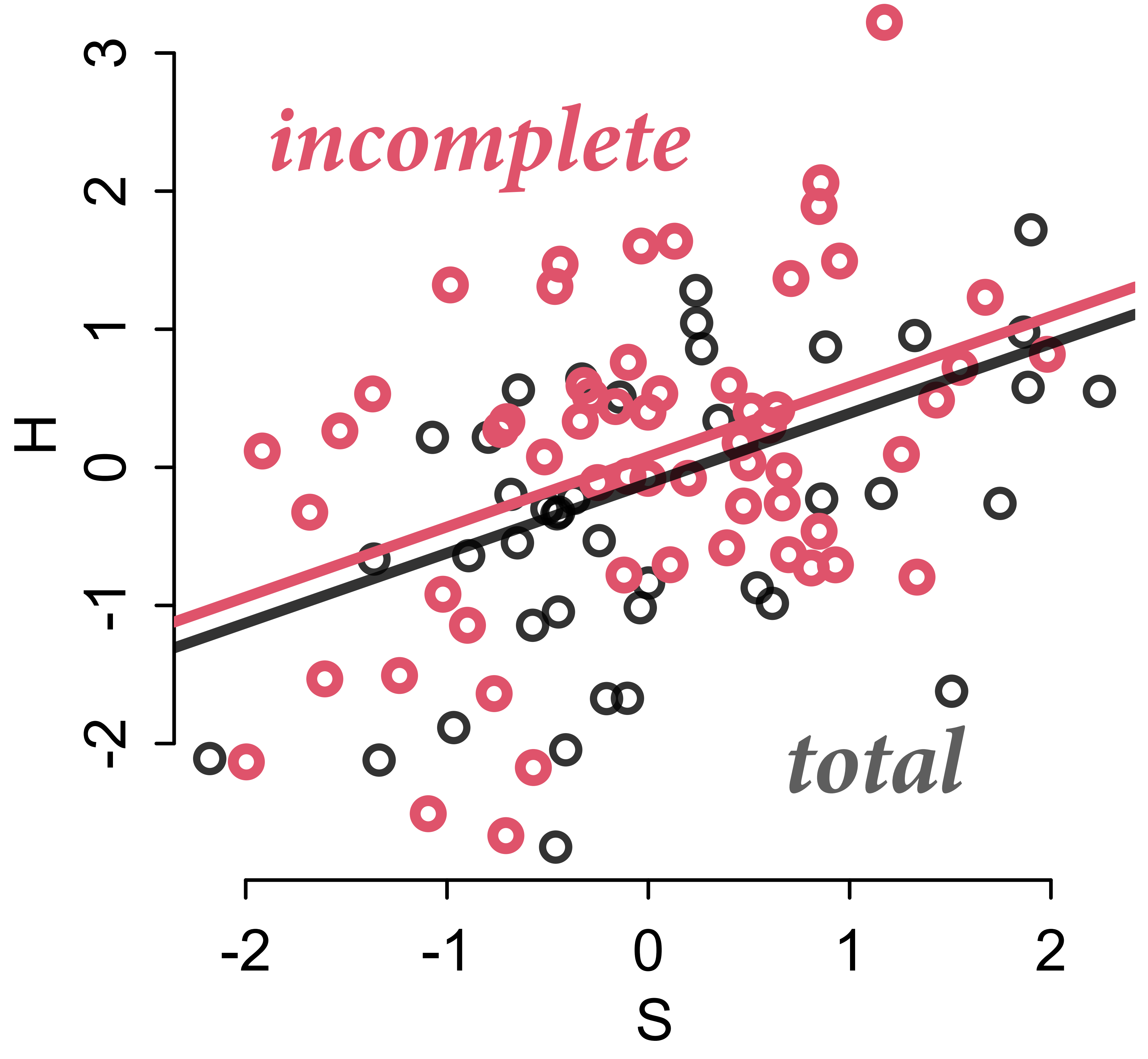


We observe the incomplete set in red

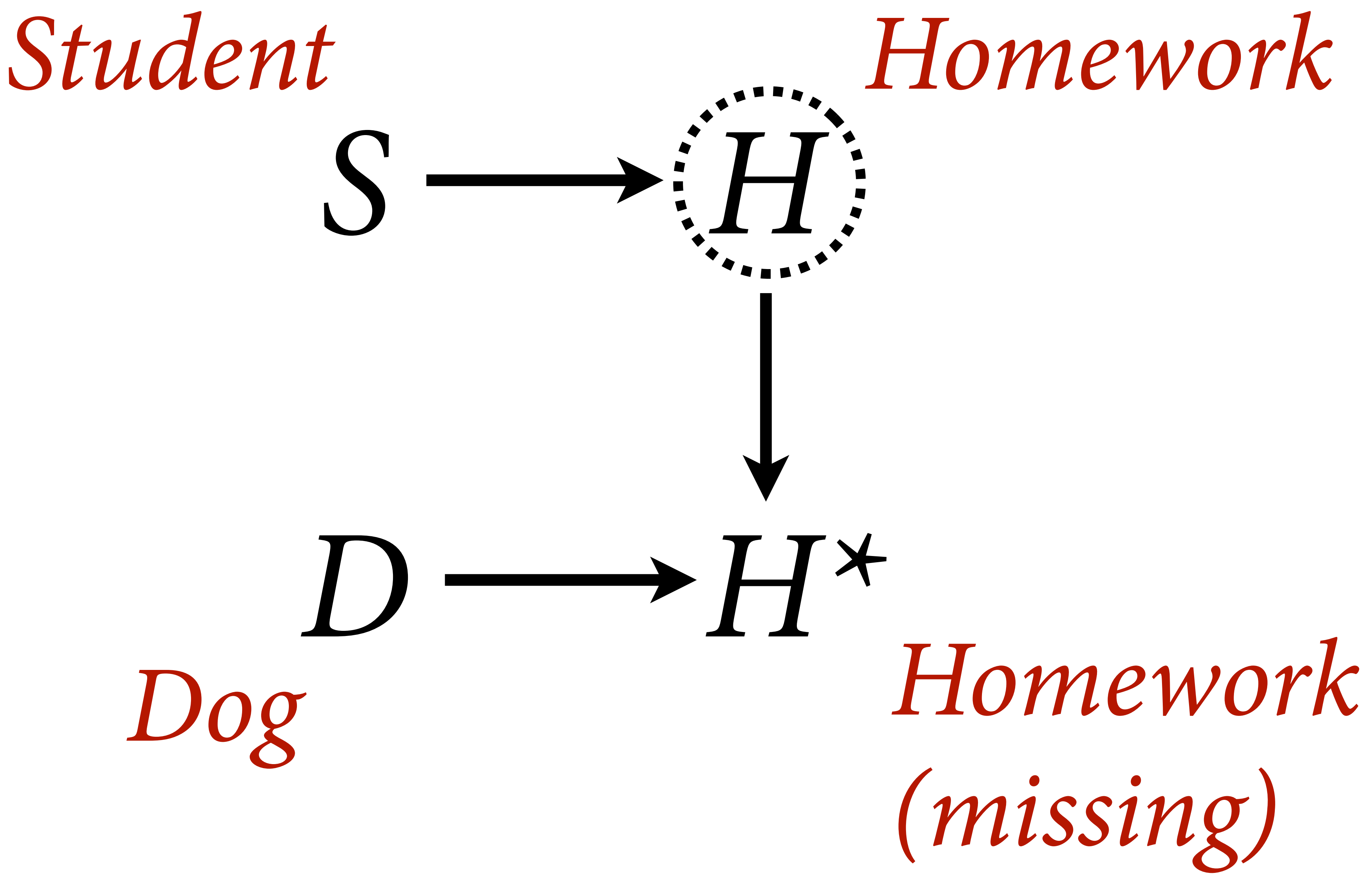
Dog usually benign

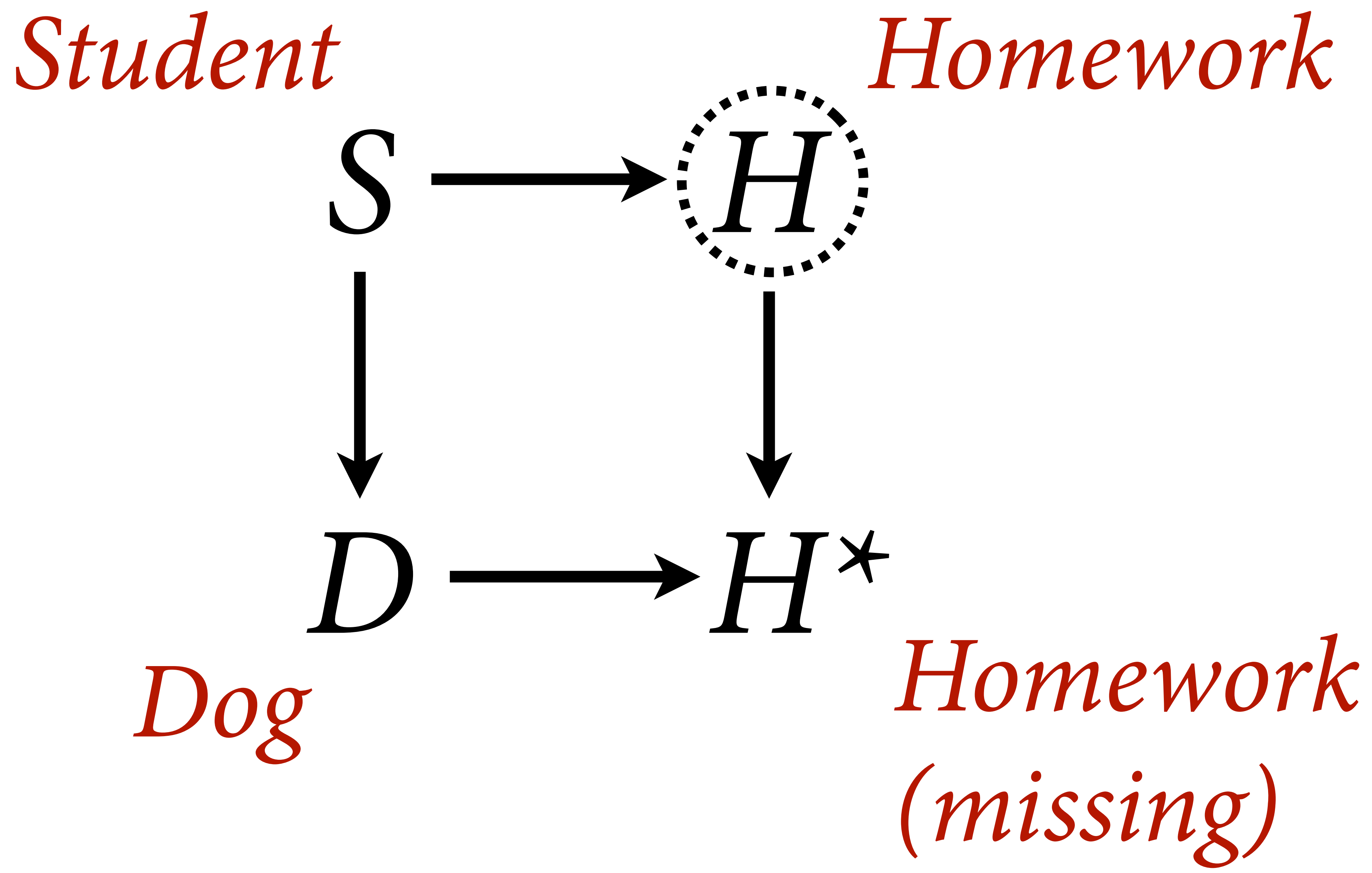
```r
# Dog eats random homework
# aka missing completely at random
N <- 100
S <- rnorm(N)
H <- rnorm(N,0.5*S)
# dog eats 50% of homework at random
D <- rbern(N,0.5)
Hstar <- H
Hstar[D==1] <- NA

plot( S , H , col=grau(0.8) , lwd=2 )
points( S , Hstar , col=2 , lwd=3 )
```

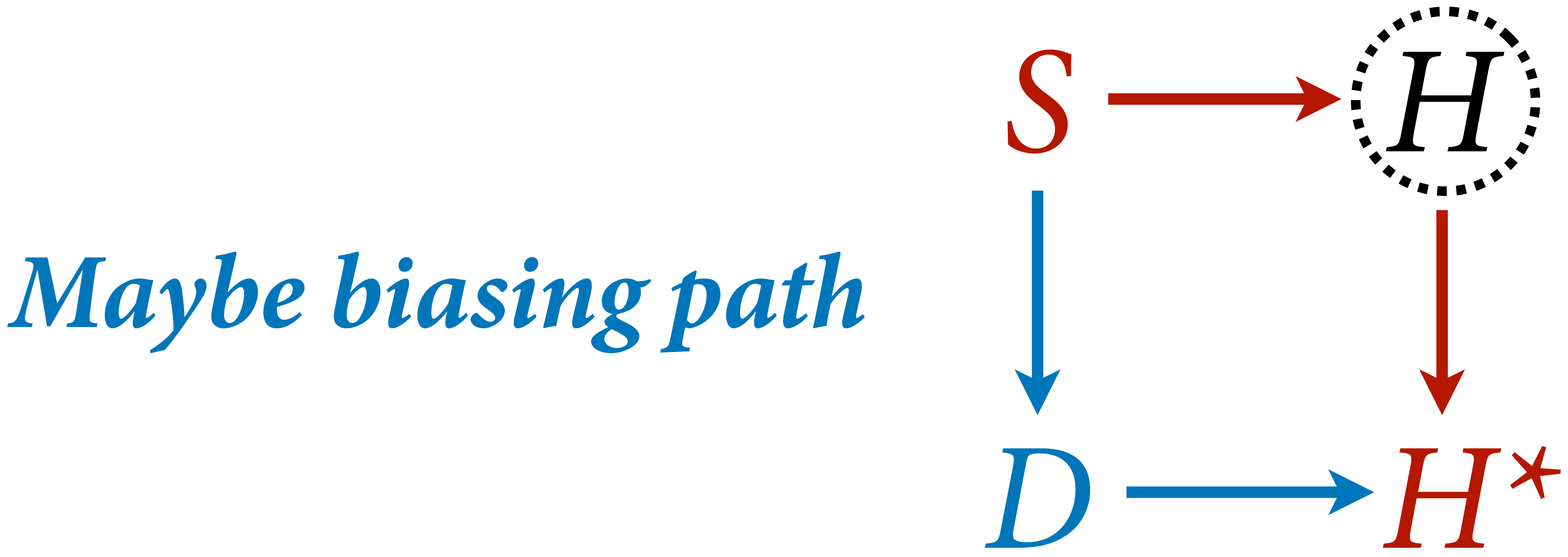




Loss of precision, usually no bias


Student
Homework
S
H
D
H*
Dog
Homework
(missing)


Student
Homework
S
H
D
H*
Dog
Homework
(missing)

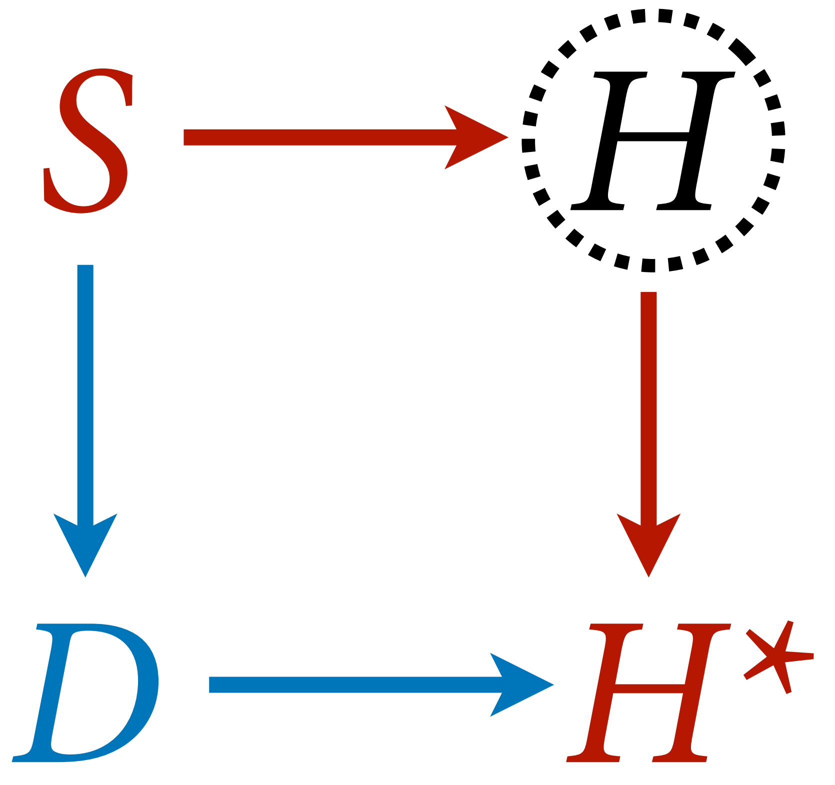


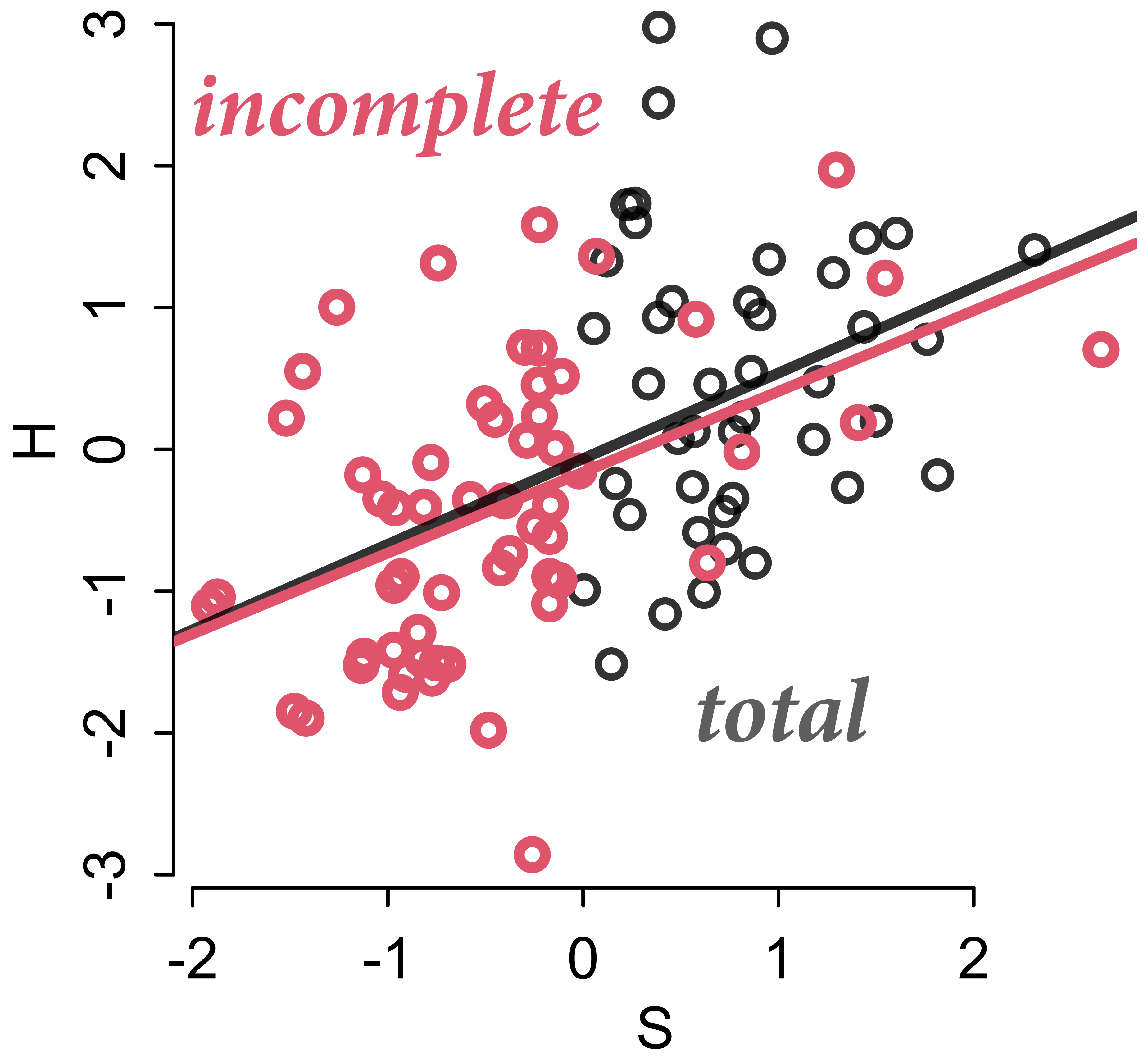
"Dog eats conditional on
cause of homework"

Missing at random (MAR)

Dog path can be benign

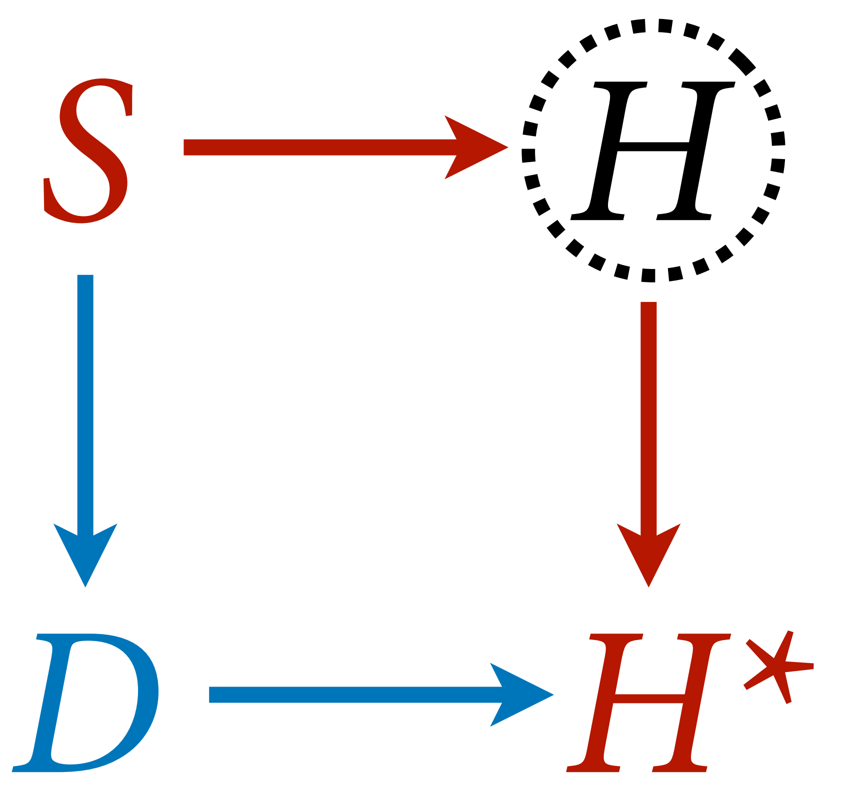
```
# Dog eats homework of students who study too much
# aka missing at random
N <- 100
S <- rnorm(N)
H <- rnorm(N,0.5*S)
# dog eats 80% of homework where S>0
D <- rbern(N, ifelse(S>0,0.8,0) )
Hstar <- H
Hstar[D==1] <- NA

plot( S , H , col=grau(0.8) , lwd=2 )
points( S , Hstar , col=2 , lwd=3 )
```

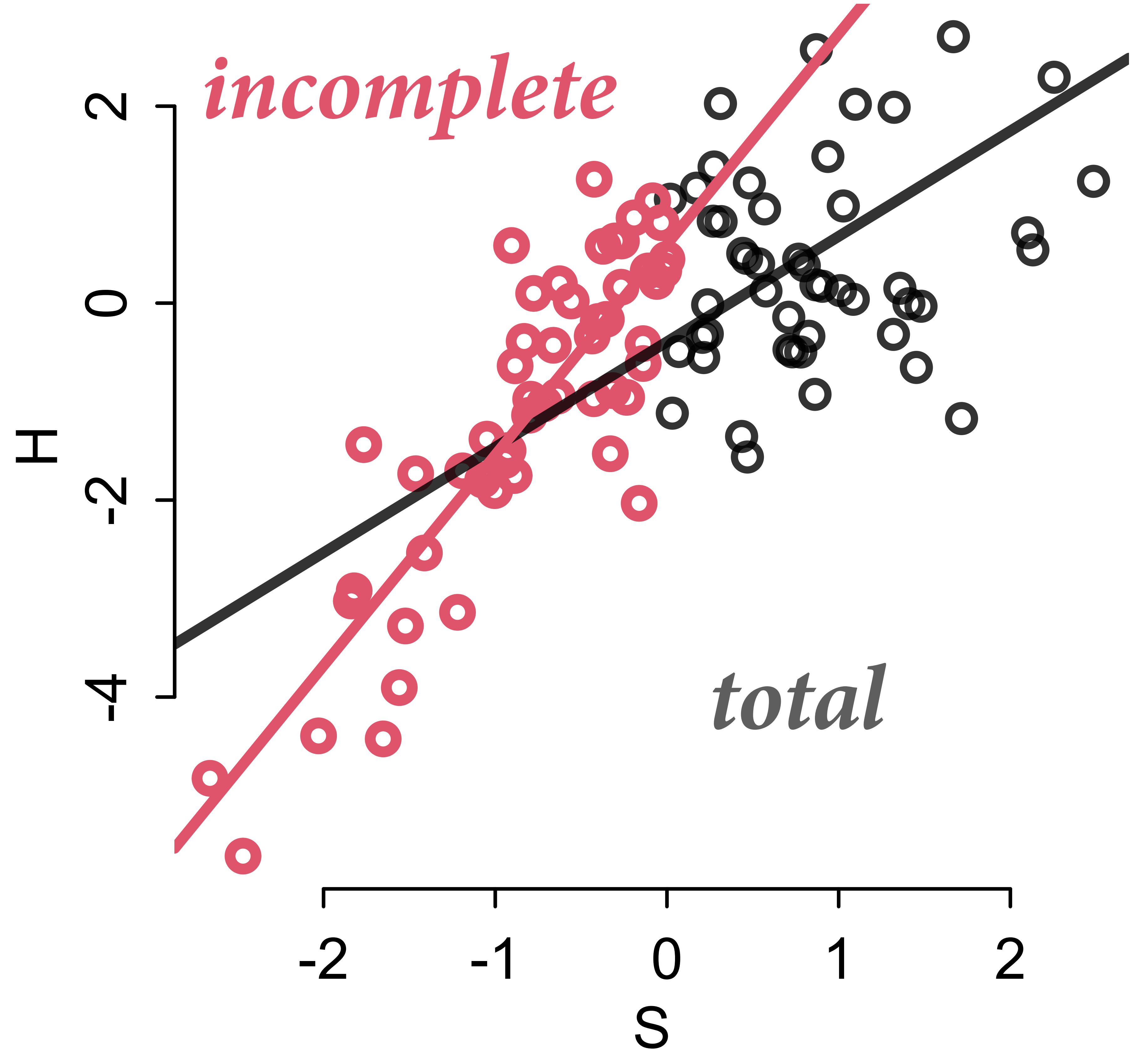


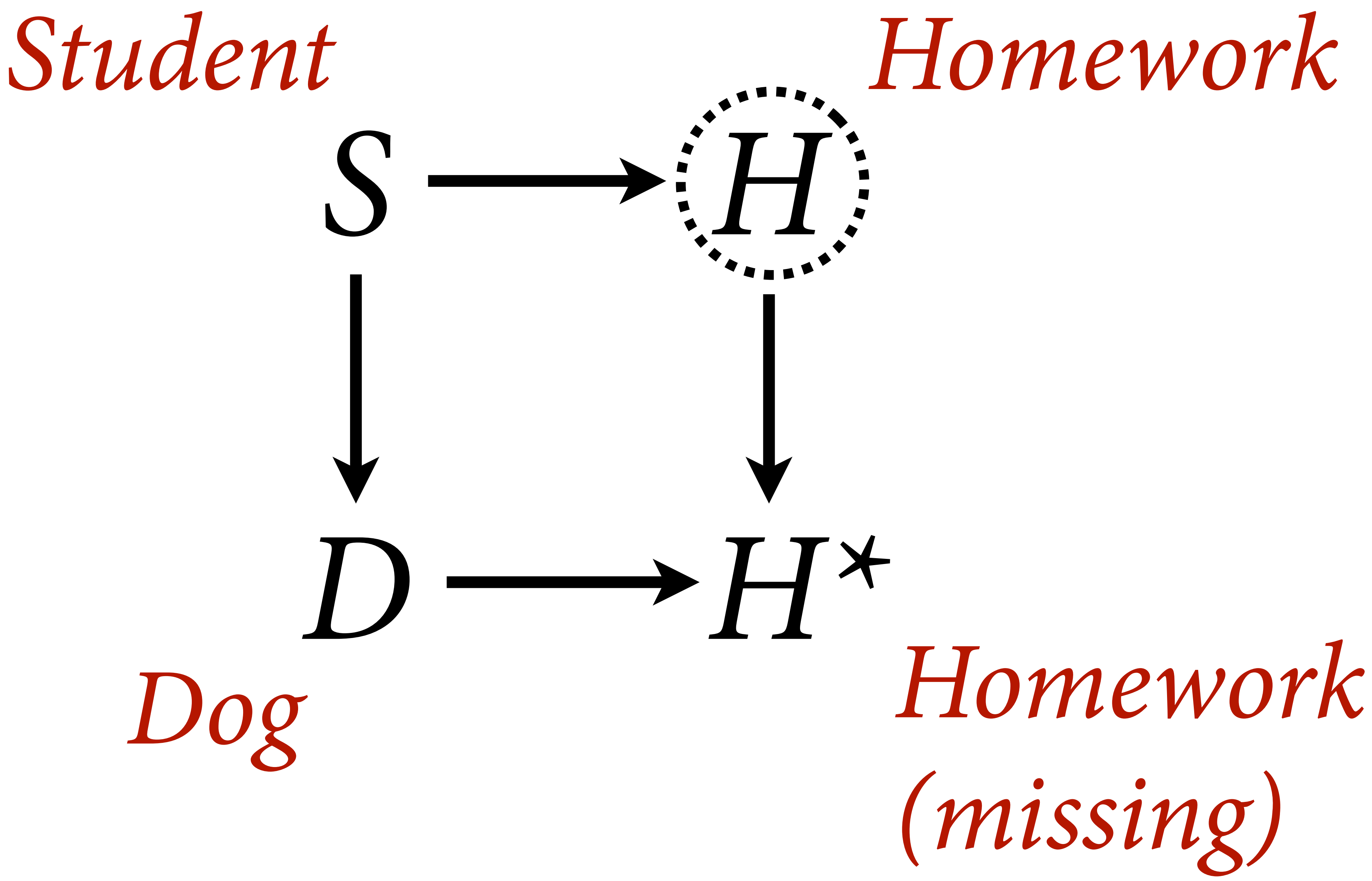


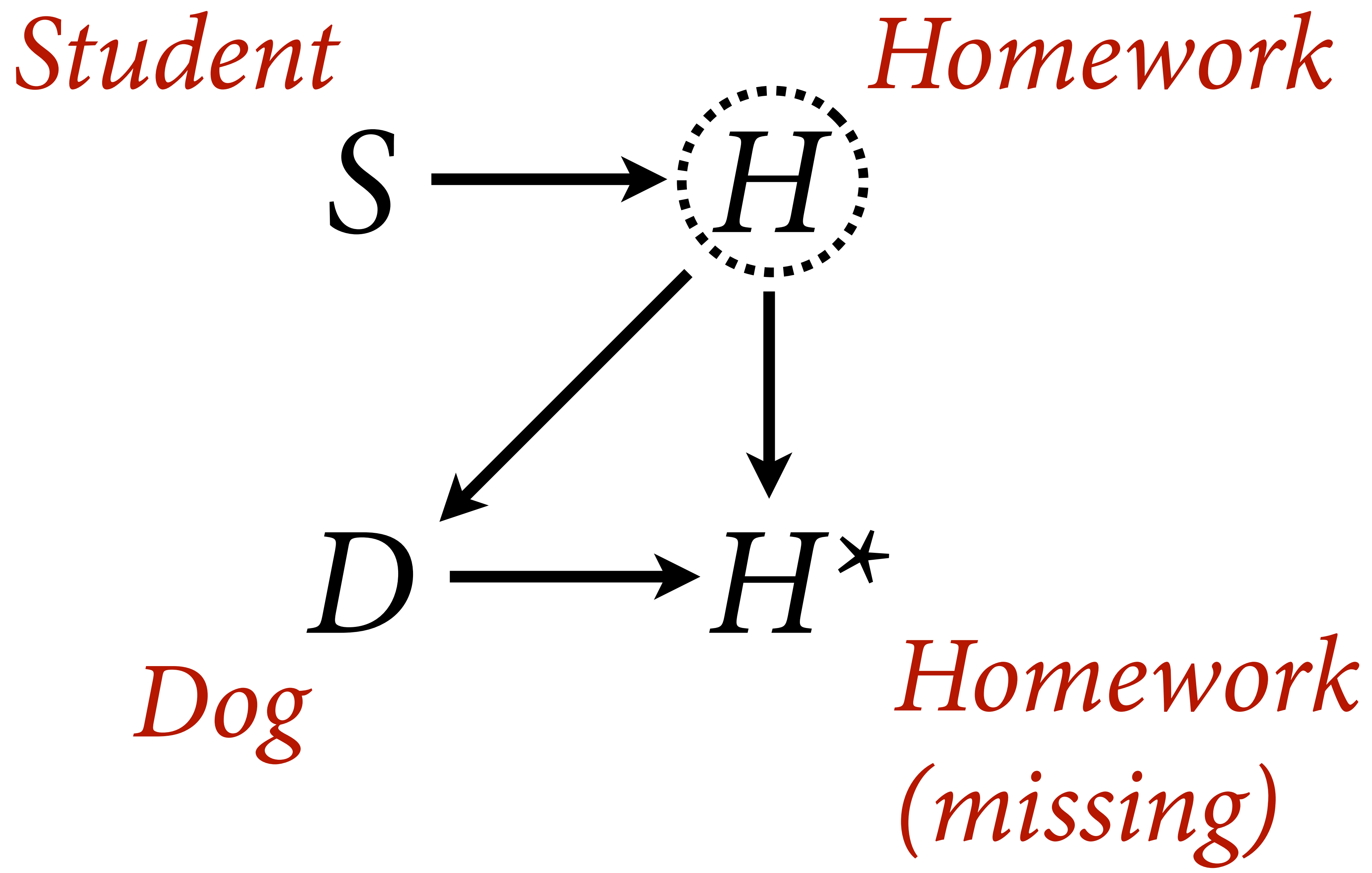
Non-linear relationships and poor modeling, less benign

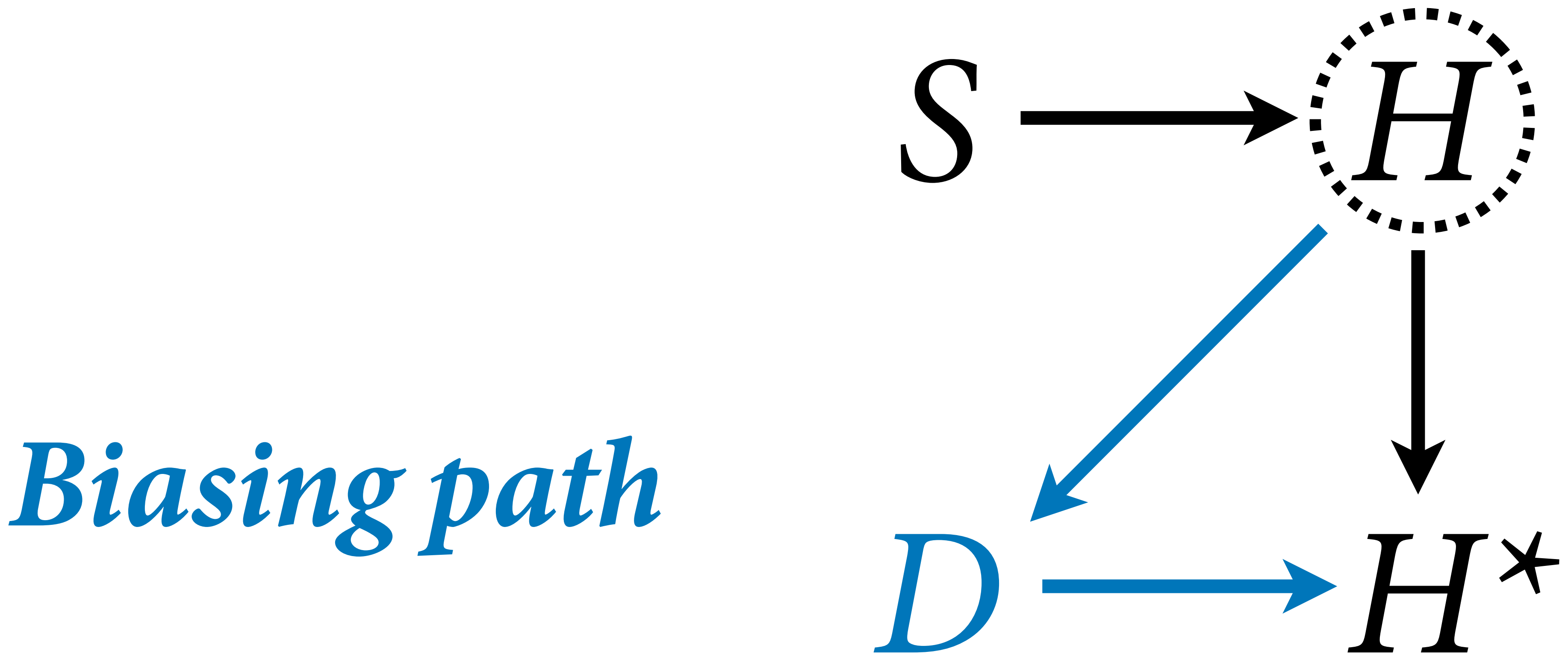
```r
# Dog eats homework of students who study too much
# BUT NOW NONLINEAR WITH CEILING EFFECT
N <- 100
S <- rnorm(N)
H <- rnorm(N,(1-exp(-0.7*S)))
# dog eats 100% of homework where S>0
D <- rbern(N, ifelse(S>0,1,0) )
Hstar <- H
Hstar[D==1] <- NA

plot( S , H , col=grau(0.8) , lwd=2 )
points( S , Hstar , col=2 , lwd=3 )
```






Student
Homework
S
H
D
H*
Dog
Homework
(missing)


Student
Homework
S
H
D
H*
Dog
Homework
(missing)

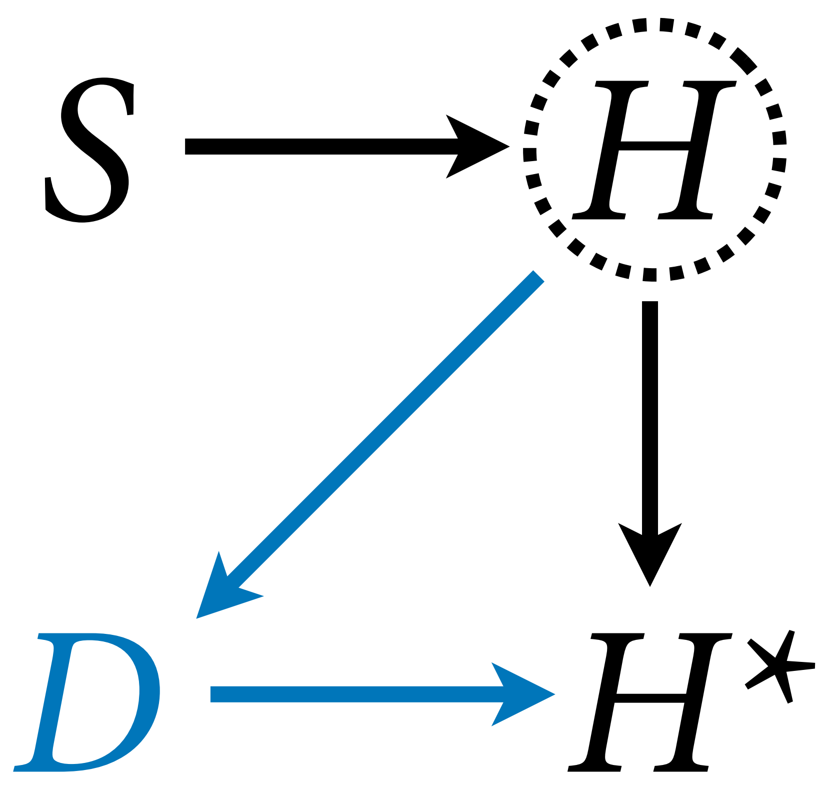


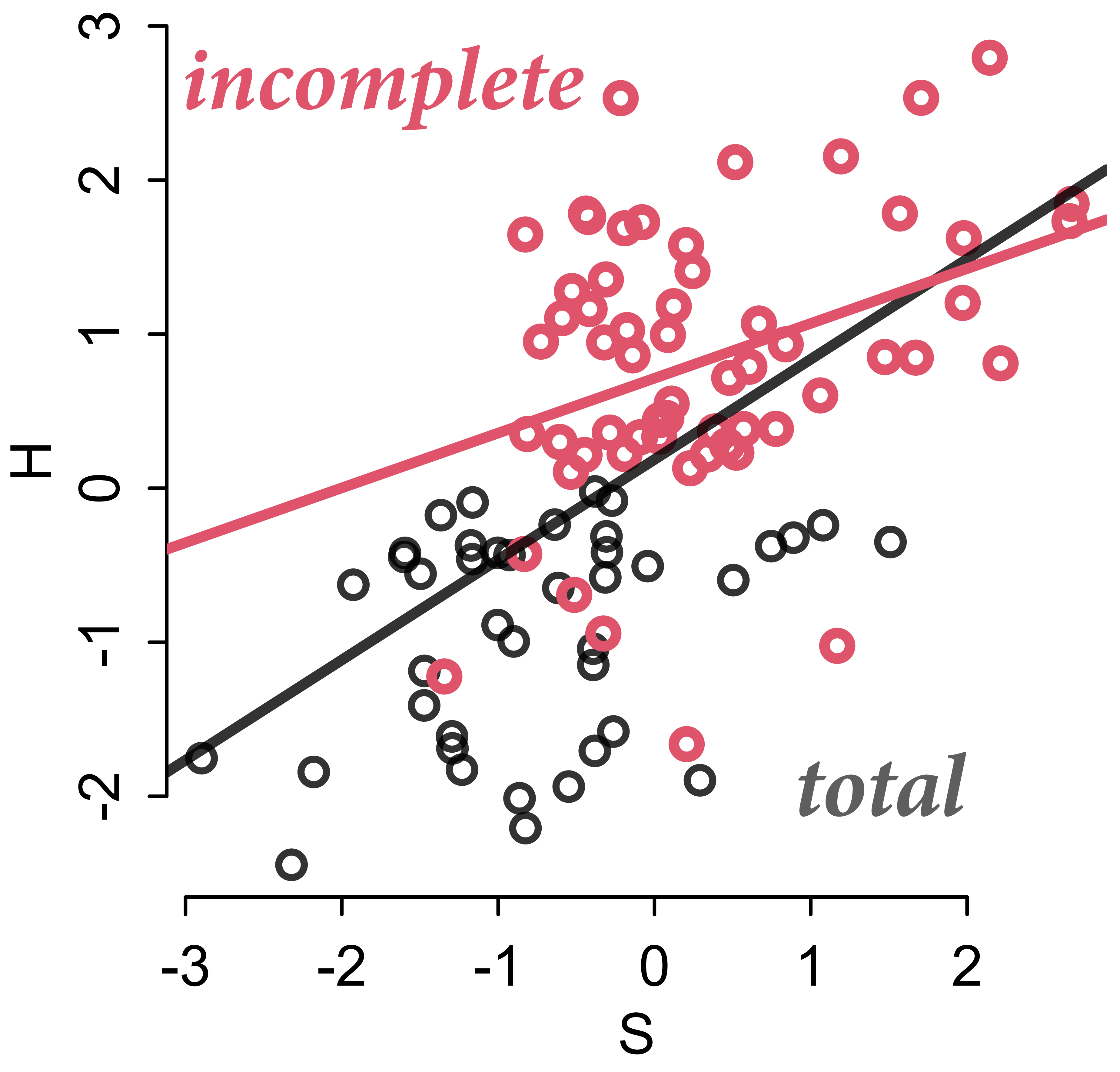
“Dog eats conditional on homework itself”

Missing not at random (MNAR)

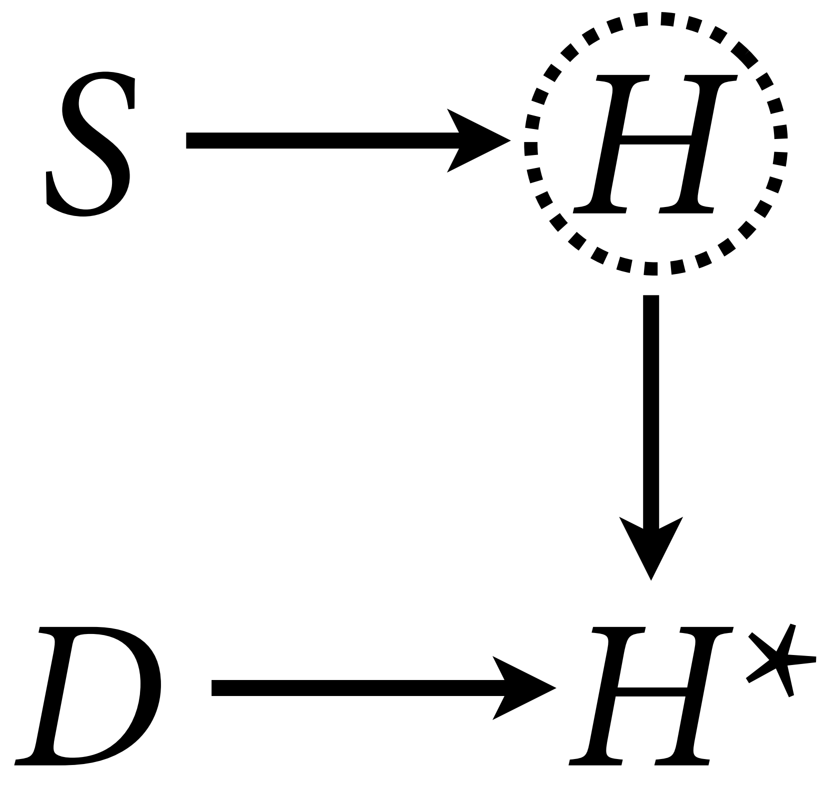
Usually not benign

```
# Dog eats bad homework
# aka missing not at random
N <- 100
S <- rnorm(N)
H <- rnorm(N,0.5*S)
# dog eats 90% of homework where H<0
D <- rbern(N, ifelse(H<0,0.9,0) )
Hstar <- H
Hstar[D==1] <- NA

plot( S , H , col=grau(0.8) , lwd=2 )
points( S , Hstar , col=2 , lwd=3 )
```

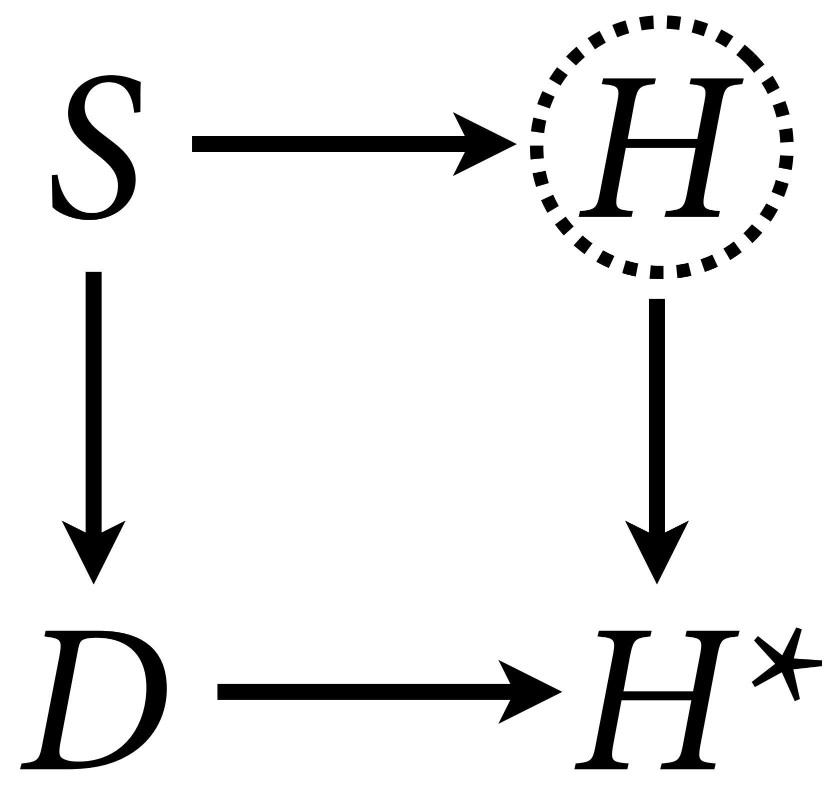




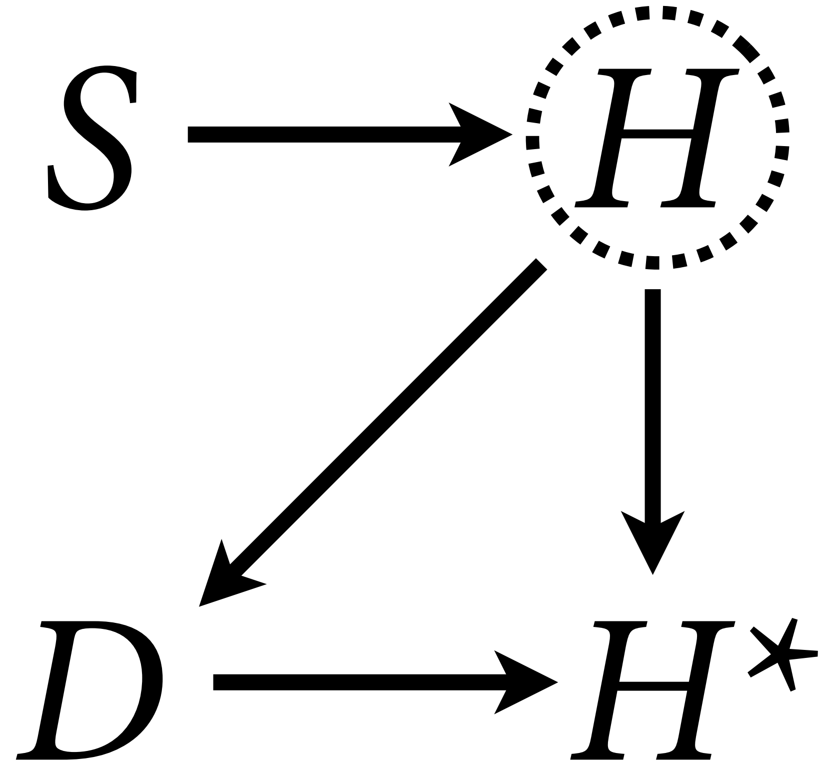
(1) Dog eats random homework: Dropping incomplete cases okay, but loss of efficiency (MCAR)



(2) Dog eats conditional on cause: Correctly condition on cause (MAR)



(3) Dog eats homework itself: Usually hopeless unless we can model the dog (e.g. survival analysis) (MNAR)



Bayesian Imputation

(1) Dog eats random homework

(2) Dog eats conditional on cause

Both imply need to impute or marginalize over missing values

Bayesian Imputation: Compute posterior probability distribution of missing values

Marginalizing unknowns: Averaging over distribution of missing values

Bayesian Imputation

Causal model of all variables implies strategy for imputation

Technical obstacles exist!

Sometimes imputation is unnecessary, e.g. discrete parameters (can use marginalization instead)

Sometimes imputation is easier, e.g. censored observations

Phylogenetic regression

data(Primates301)

Life history traits

Mass g, brain cc, group size

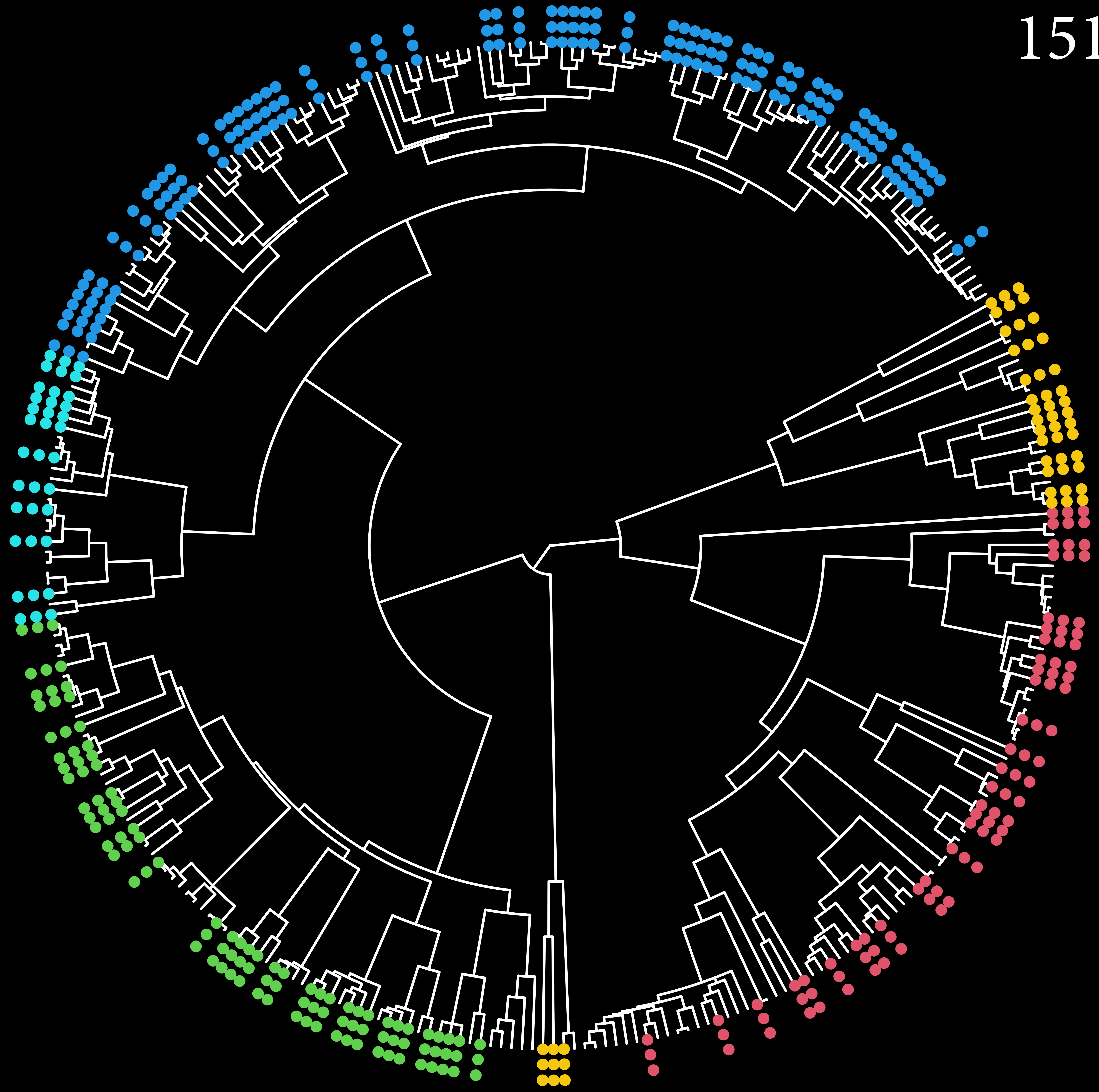
Much missing data, measurement error, unobserved confounding

Street et al 2017 Coevolution of cultural intelligence, extended life history, sociality, and brain size in primates

301 species

301 species

151 complete cases



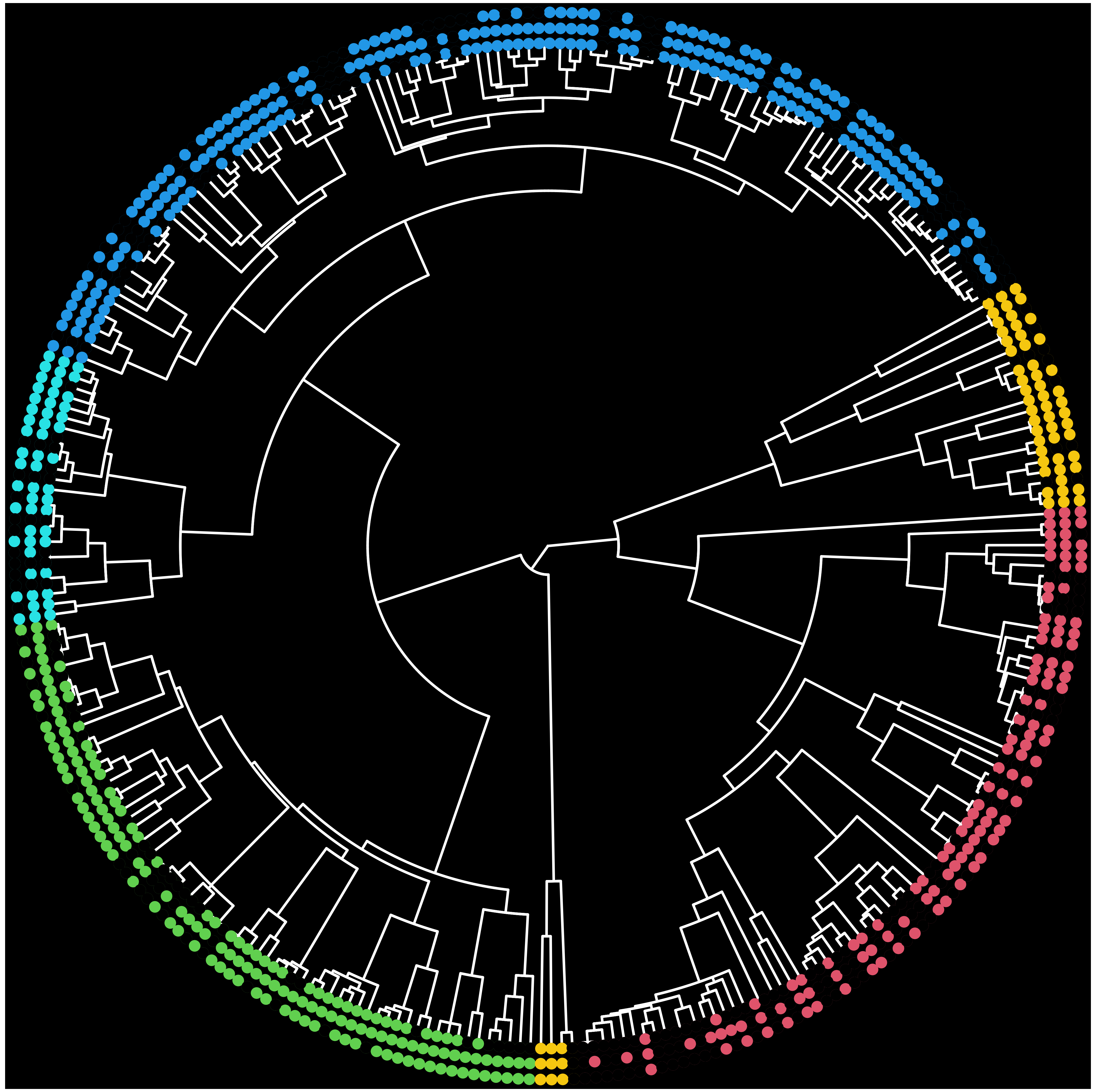
Imputing Primates

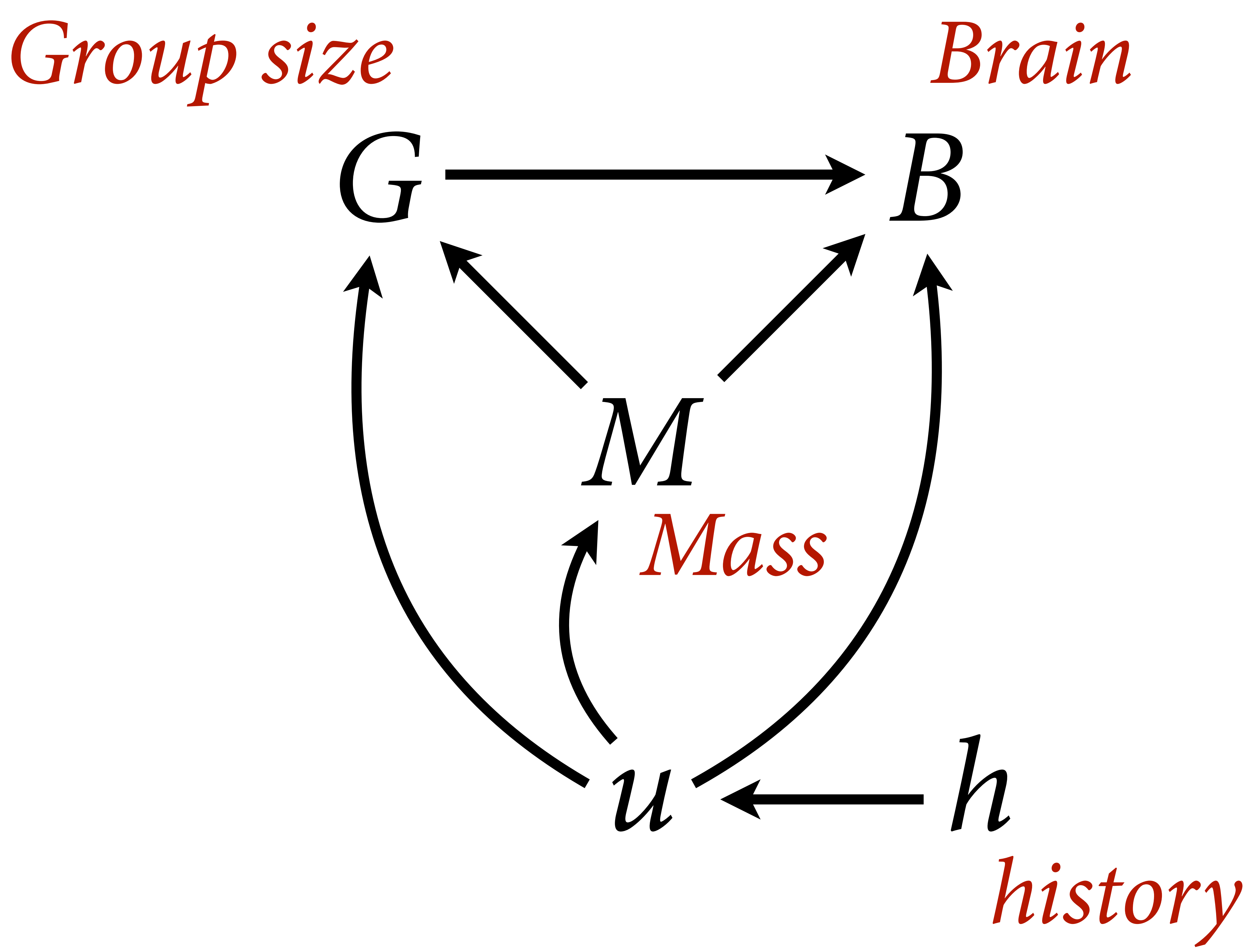
Key idea: Missing values already have probability distributions

Express causal model for each partially-observed variable

Replace each missing value with a parameter, let model do the rest

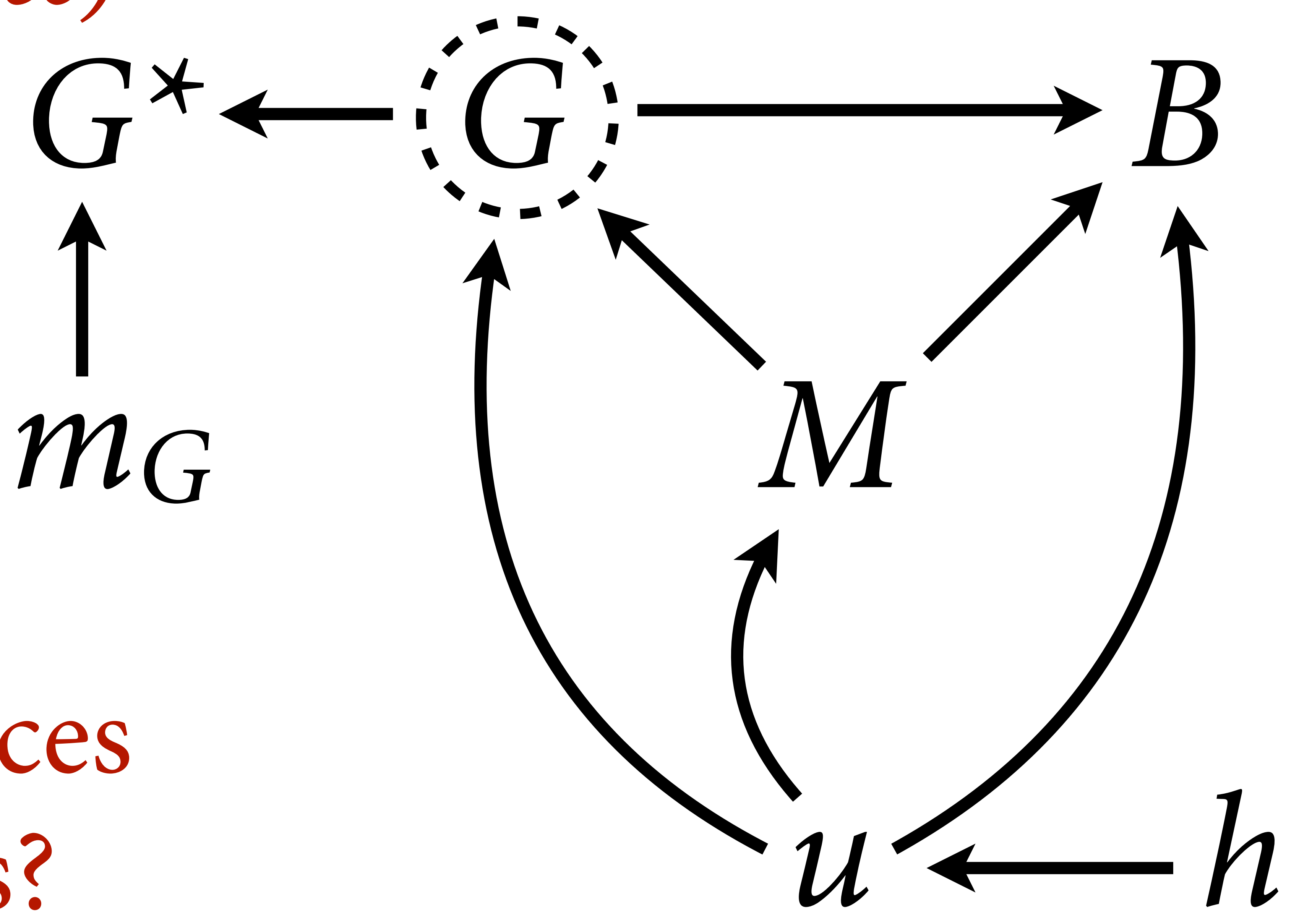
Conceptually weird, technically awkward



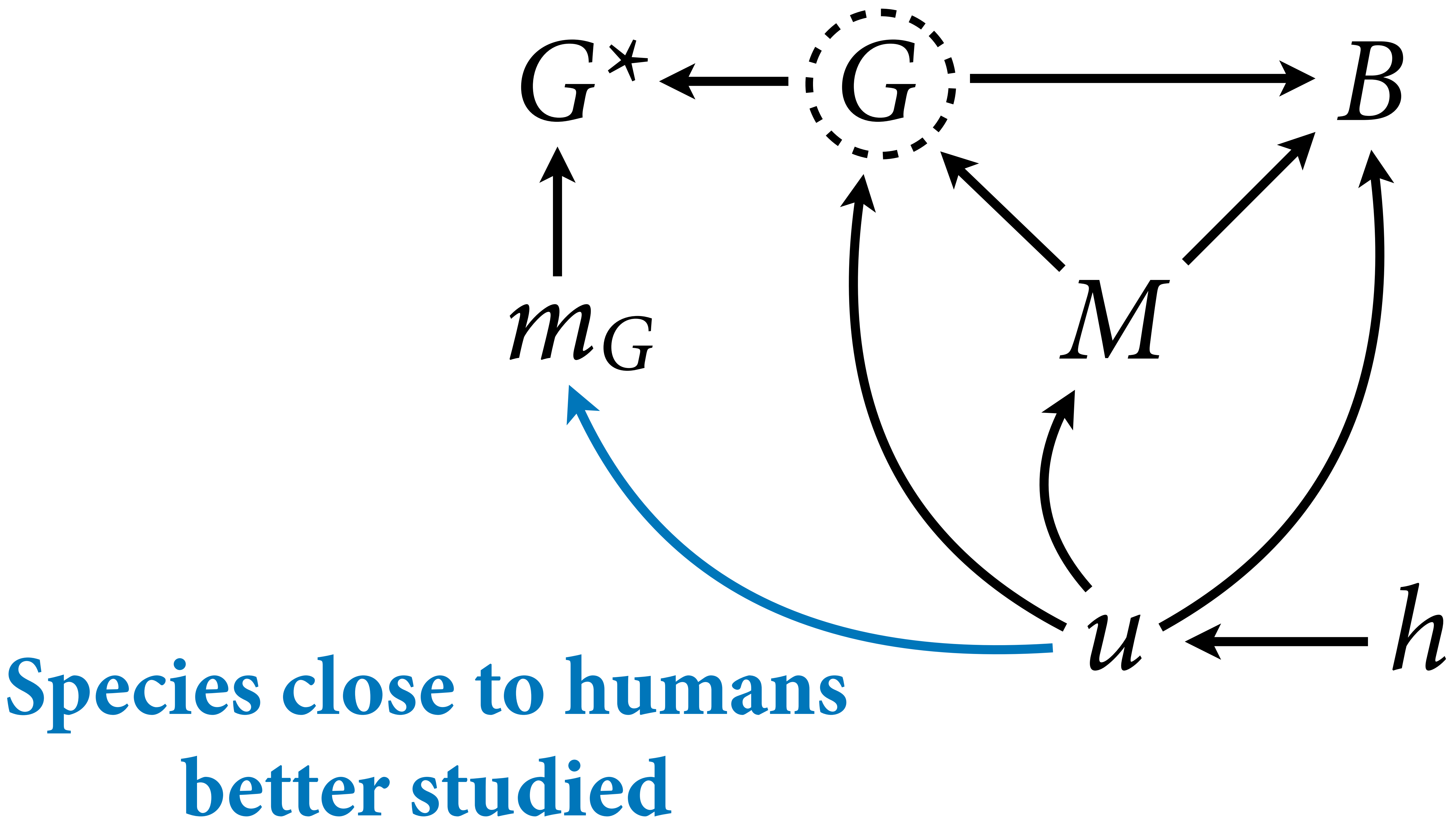


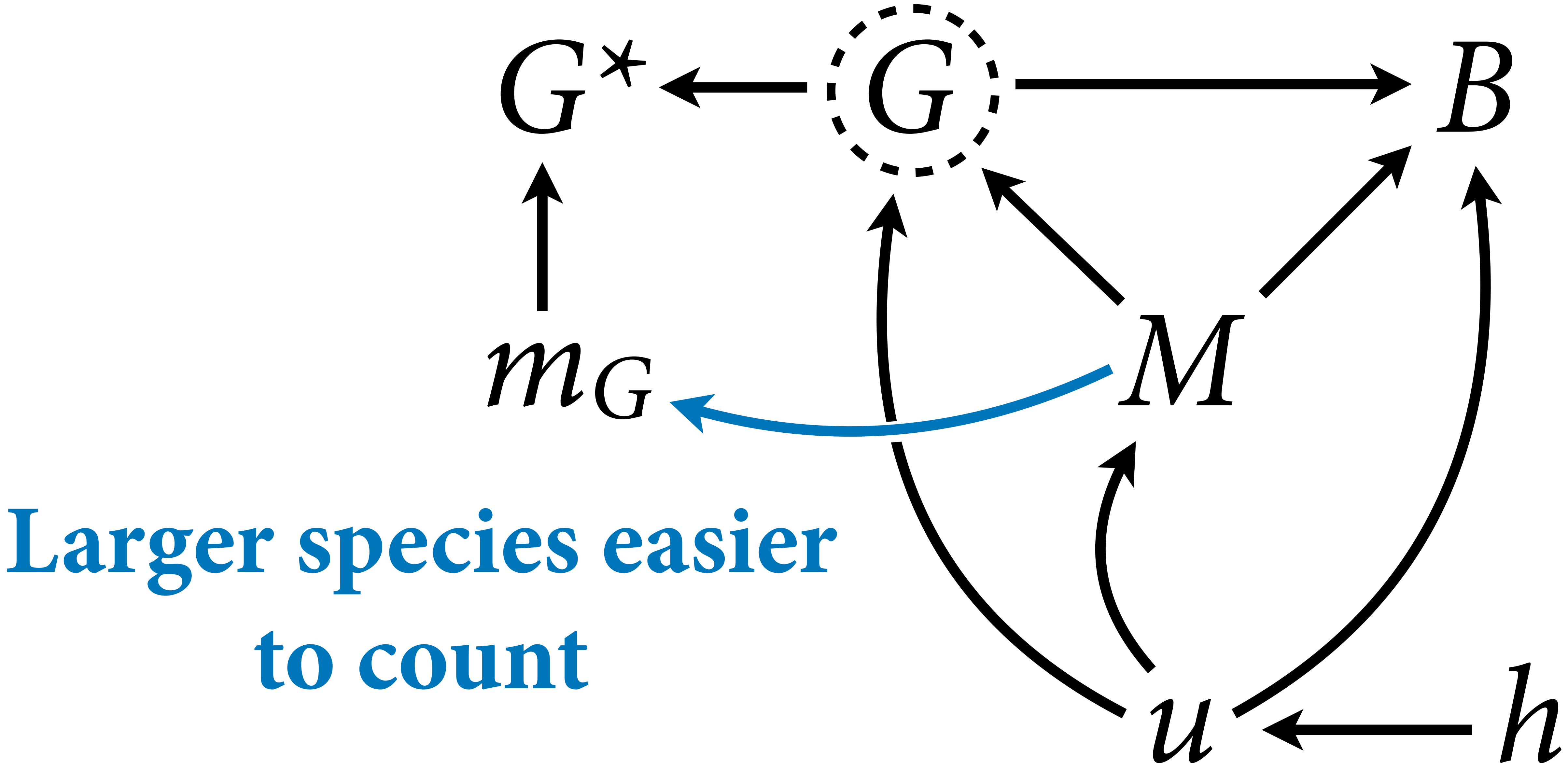
Group size
Brain
G
B
M
Mass
u
h
history

Group size
(missing values)

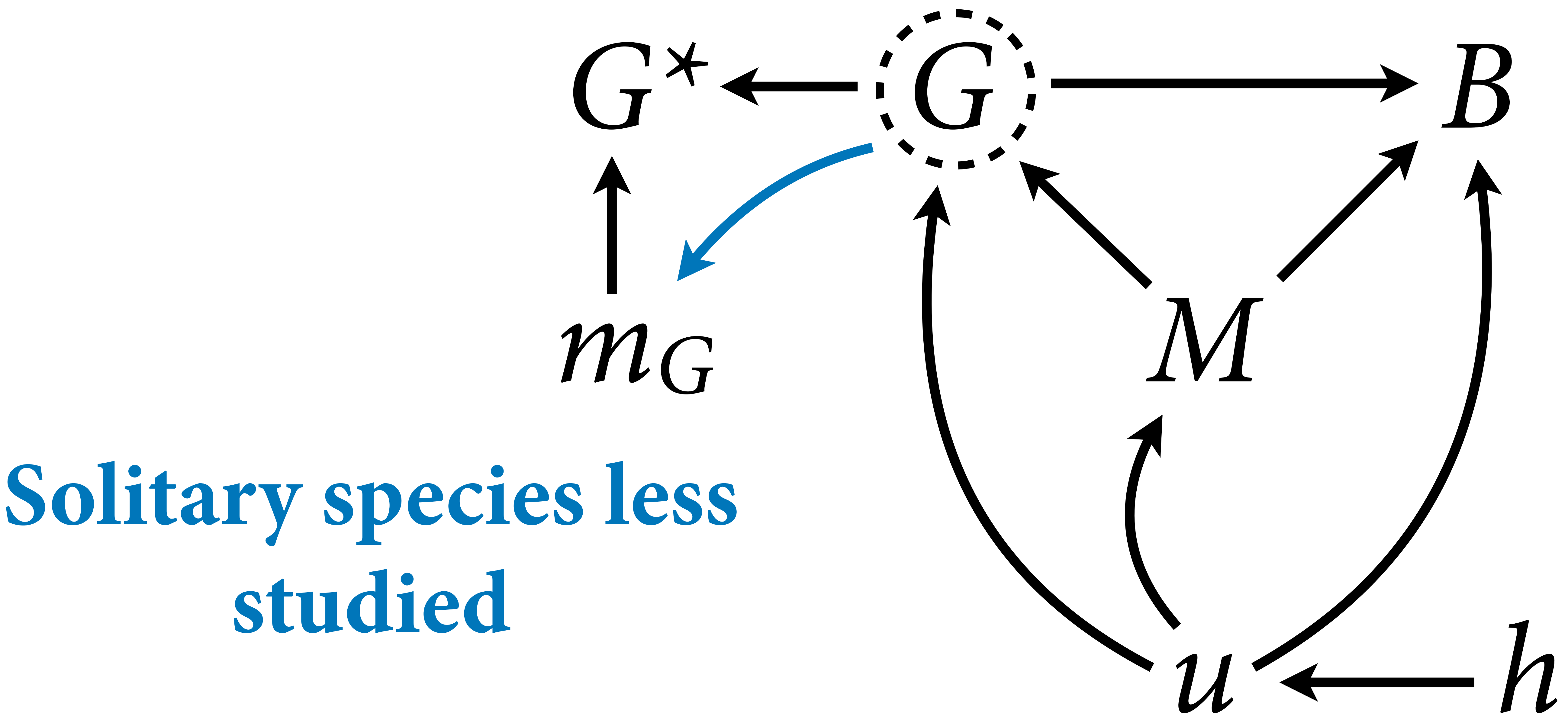


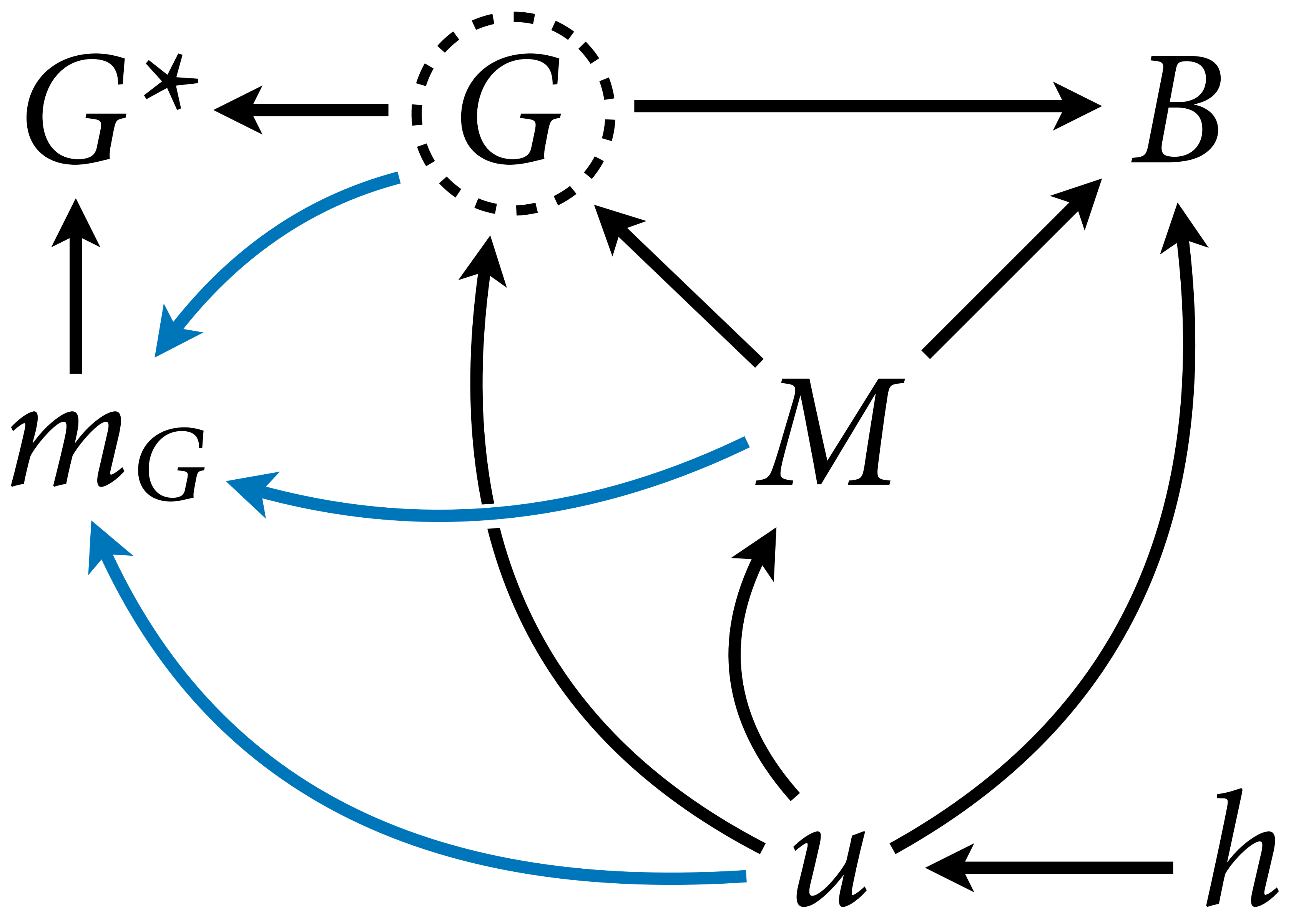
What influences
missingness?


G*
G
B
M
m_G
u
h
Species close to humans
better studied



G^*
G
B
m_G
M
Larger species easier to count
u
h

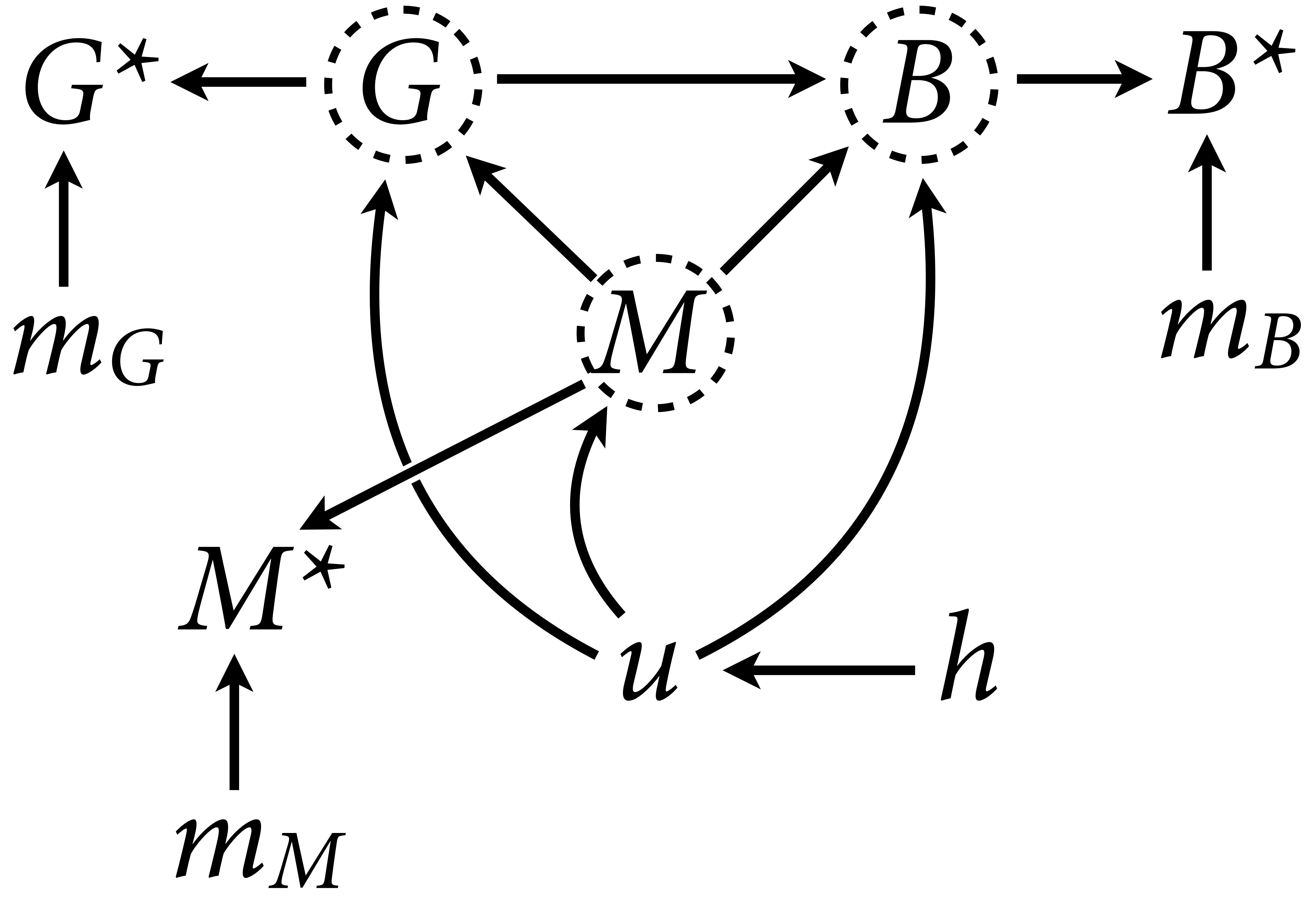

G*
G
B
m_G
M
u
h
Solitary species less studied



Whatever the assumption, our goal is to use the causal model to infer probability distribution of each missing value.

Uncertainty in each missing value cascades through the entire model.

The uncertainty about specific values can lead to more (or less) precise estimates



G^*
G
B
B^*
m_G
M
m_B
M^*
u
h
m_M

$$B \sim \text{MVNormal}(\mu, \mathbf{K})$$

$$\mu_i = \alpha + \beta_G G_i + \beta_M M_i$$

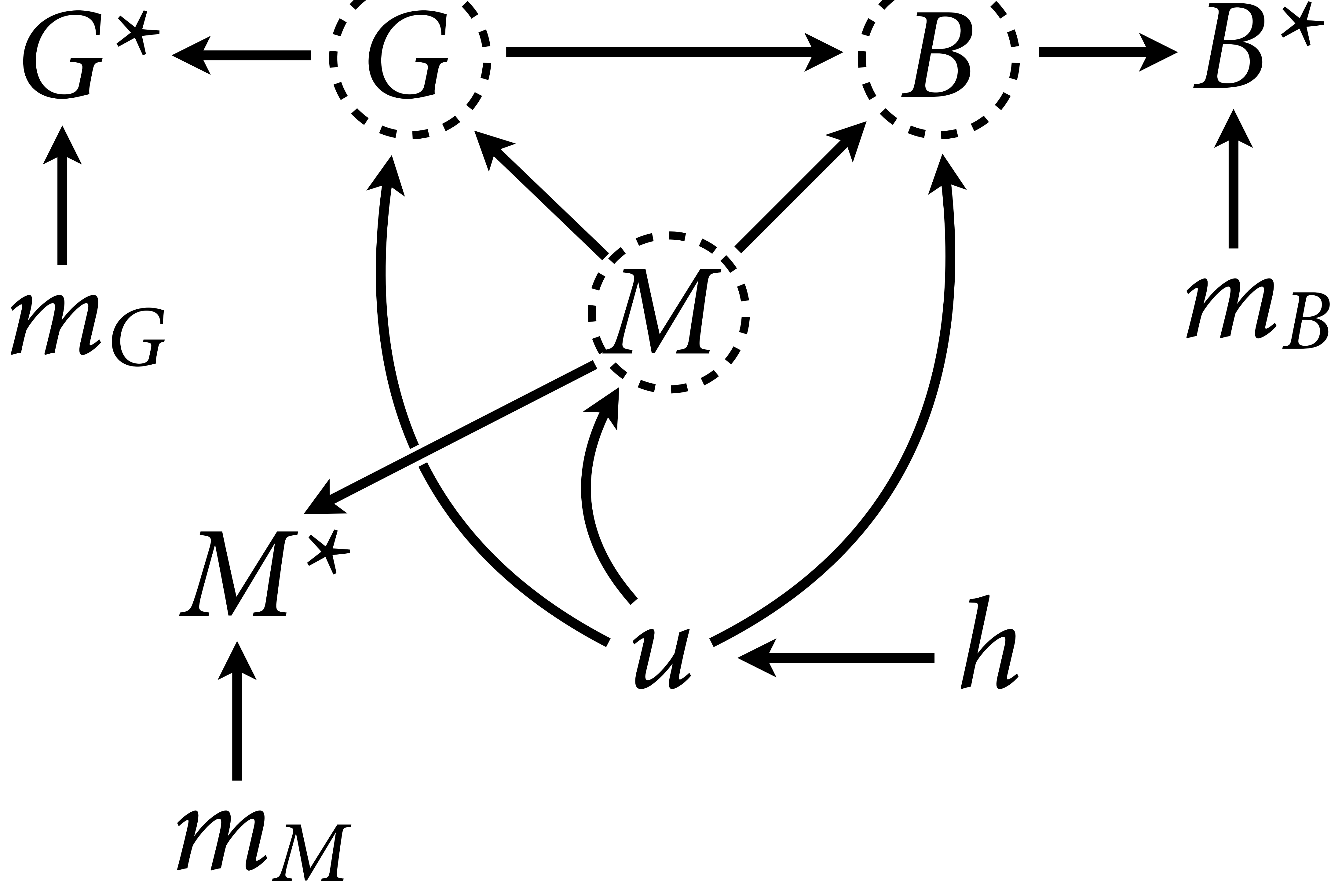
$$\mathbf{K} = \eta^2 \exp(-\rho d_{i,j})$$

$$\alpha \sim \text{Normal}(0,1)$$

$$\beta_G, \beta_M \sim \text{Normal}(0,0.5)$$

$$\eta^2 \sim \text{HalfNormal}(1,0.25)$$

$$\rho \sim \text{HalfNormal}(3,0.25)$$



$$
\begin{tikzcd}
G \arrow[rrr] & & & B \\
& M \arrow[urr] & & \\
& & h \arrow[r] & u \arrow[uu, bend right]
\end{tikzcd}
$$

$$
\begin{aligned}
B &\sim \text{MVNormal}(\mu, \mathbf{K}) \\
\mu_i &= \alpha + \beta_G G_i + \beta_M M_i \\
\mathbf{K} &= \eta^2 \exp(-\rho d_{i,j}) \\
\alpha &\sim \text{Normal}(0,1) \\
\beta_G, \beta_M &\sim \text{Normal}(0,0.5) \\
\eta^2 &\sim \text{HalfNormal}(1,0.25) \\
\rho &\sim \text{HalfNormal}(3,0.25)
\end{aligned}
$$

$$\begin{tikzcd} G \arrow[rr] & & B \\ M \arrow[urr] & & \\ & h \arrow[r] & u \arrow[uu, bend right] \end{tikzcd}$$

$$\begin{aligned}
B &\sim \text{MVNormal}(\mu, \mathbf{K}) \\
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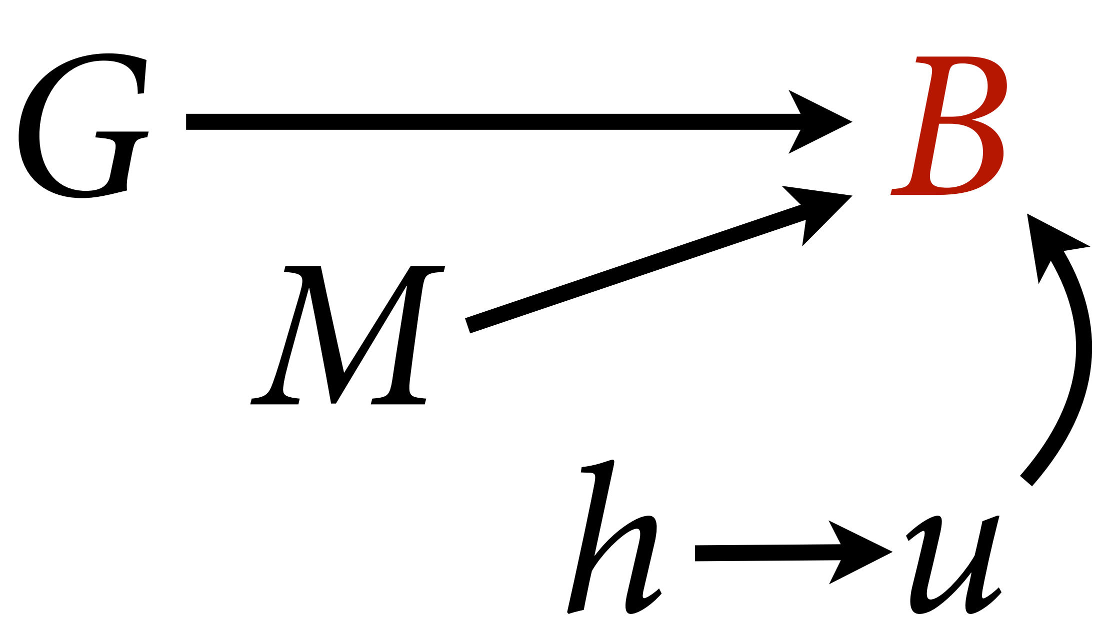
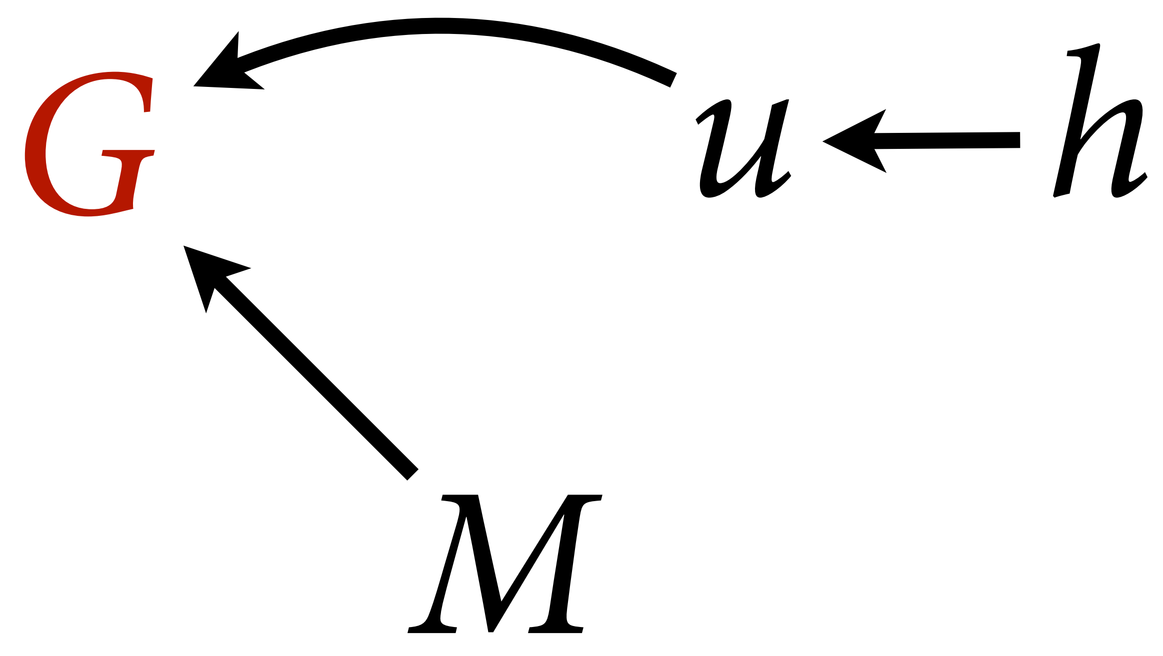
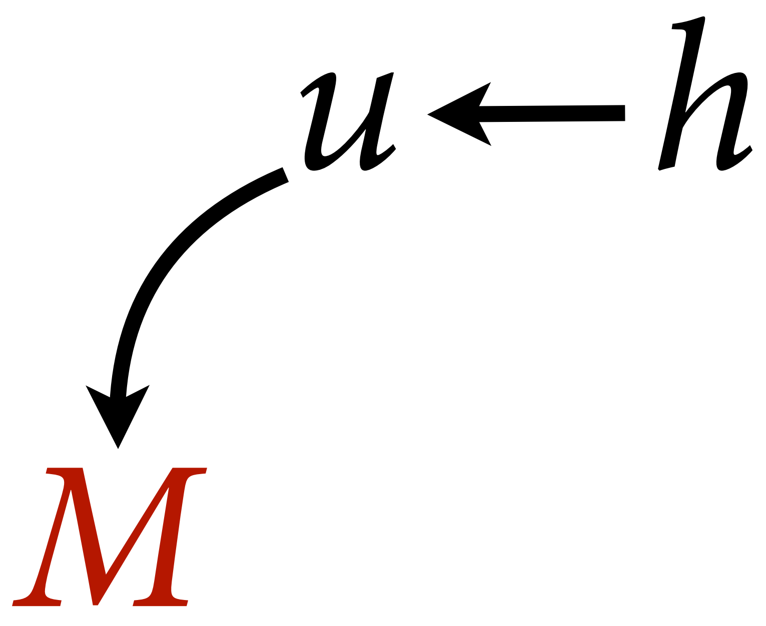
\eta^2 &\sim \text{HalfNormal}(1,0.25) \\
\rho &\sim \text{HalfNormal}(3,0.25)
\end{aligned}$$

$$\begin{tikzcd} G & u \arrow[l, bend right] & h \arrow[l] \\ & M \arrow[ul] & \end{tikzcd}$$

$$\begin{aligned}
G &\sim \text{MVNormal}(\nu, \mathbf{K}_G) \\
\nu_i &= \alpha_G + \beta_{MG} M_i \\
\mathbf{K}_G &= \eta_G^2 \exp(-\rho_G d_{i,j}) \\
\alpha_G &\sim \text{Normal}(0,1) \\
\beta_{MG} &\sim \text{Normal}(0,0.5) \\
\eta_G^2 &\sim \text{HalfNormal}(1,0.25) \\
\rho_G &\sim \text{HalfNormal}(3,0.25)
\end{aligned}$$







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$$\alpha_G \sim \text{Normal}(0,1)$$
$$\beta_{MG} \sim \text{Normal}(0,0.5)$$
$$\eta_G^2 \sim \text{HalfNormal}(1,0.25)$$
$$\rho_G \sim \text{HalfNormal}(3,0.25)$$

$$M \sim \text{MVNormal}(0,\mathbf{K}_M)$$
$$\mathbf{K}_M = \eta_M^2 \exp(-\rho_M d_{i,j})$$
$$\eta_M^2 \sim \text{HalfNormal}(1,0.25)$$
$$\rho_M \sim \text{HalfNormal}(3,0.25)$$

Draw the Missing Owl

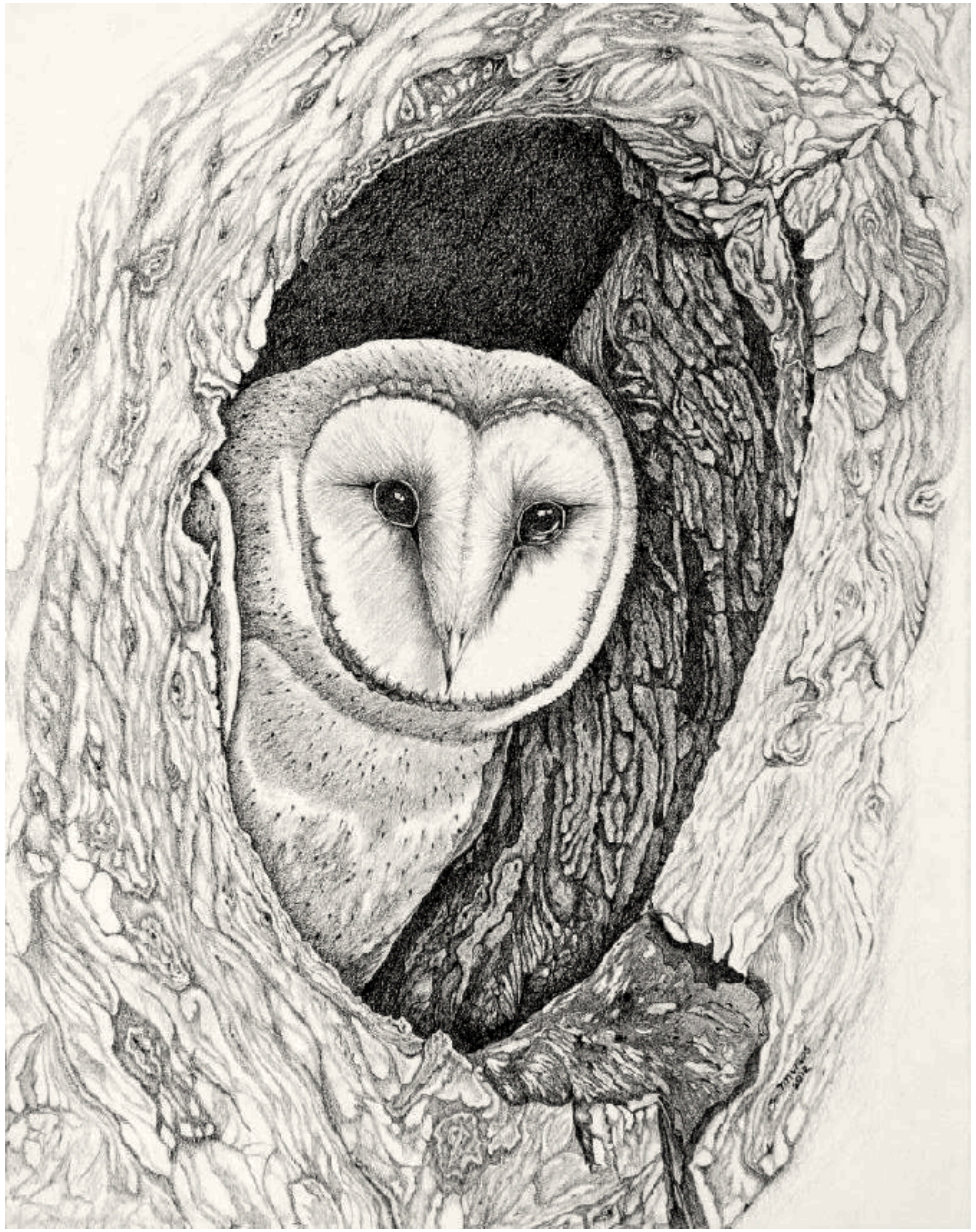
Let's take it slow…

(1) Ignore cases with missing B values (for now)

(2) Impute G and M ignoring models for each

(3) Impute G using model

(4) Impute B, G, M using model



Draw the Missing Owl

Let's take it slow…

(1) Ignore cases with missing *B* values (for now)

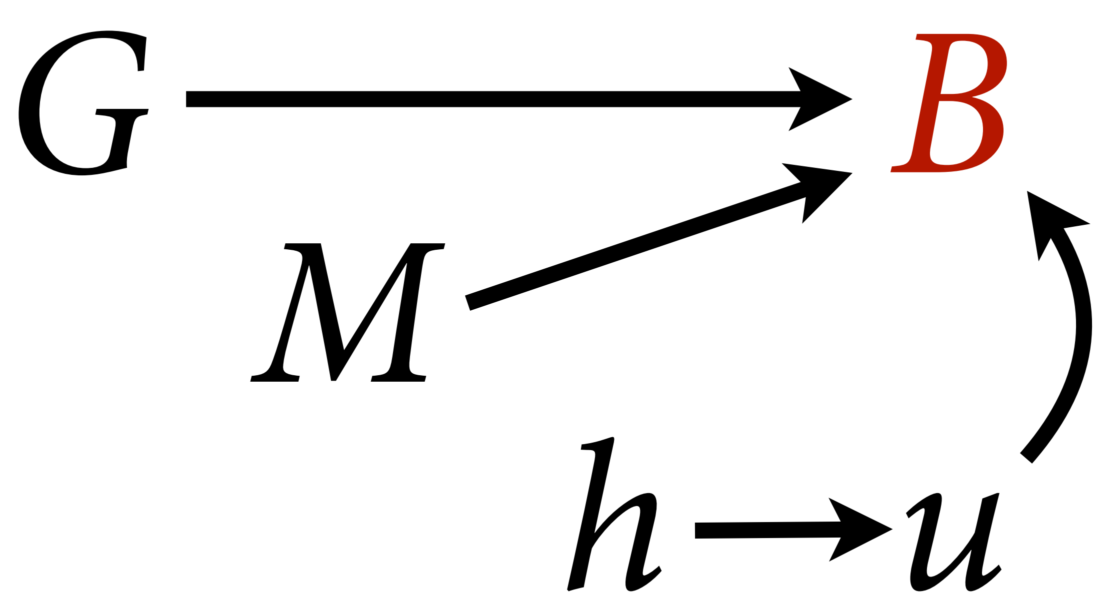
```
> dd <- d[complete.cases(d$brain),]
> table( M=!is.na(dd$body) , G=!is.na(dd$group_size) )
       G
M       FALSE TRUE
  FALSE     2    0
  TRUE     31  151
```

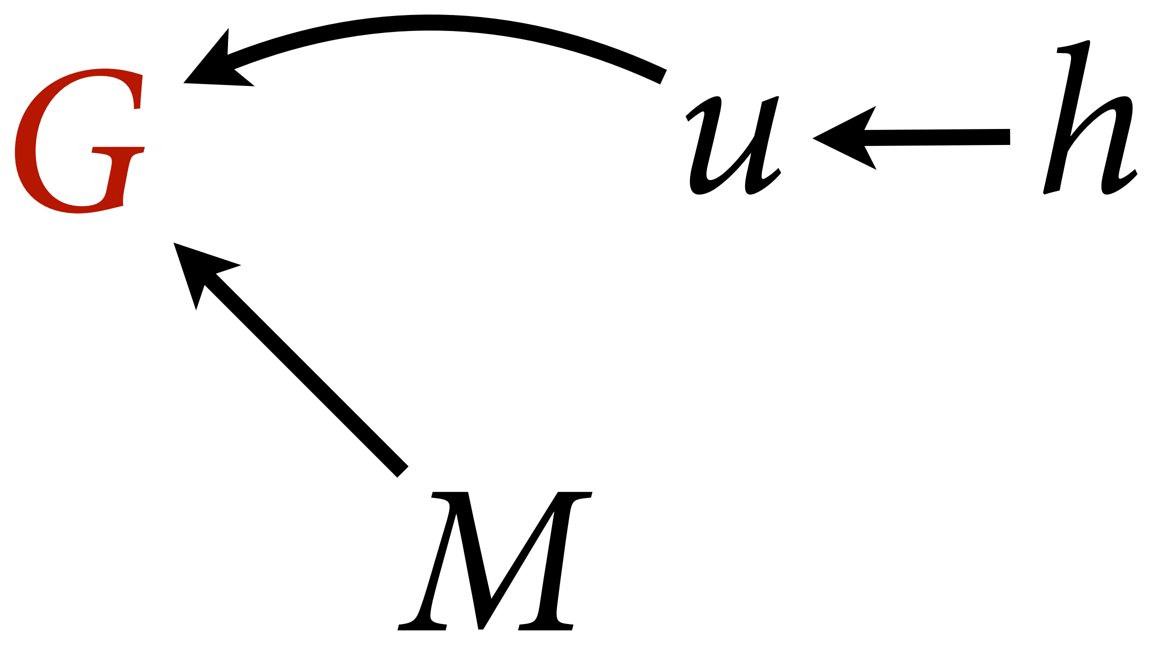
(2) Impute *G* and *M* ig
for each

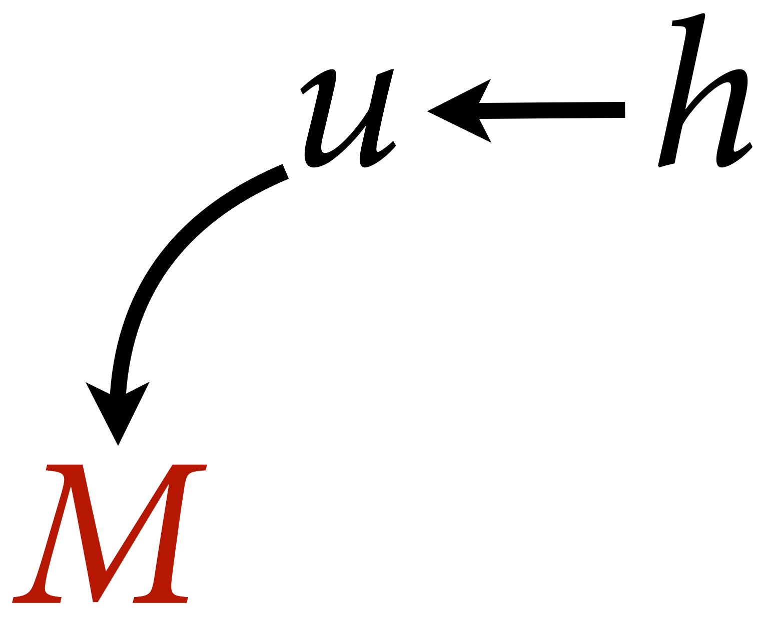
(3) Impute *G* using model

(4) Impute *B*, *G*, *M* using model

(2) Impute G and M ignoring models for each







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$$\mathbf{K} = \eta^2 \exp(-\rho d_{i,j})$$
$$\alpha \sim \text{Normal}(0,1)$$
$$\beta_G, \beta_M \sim \text{Normal}(0,0.5)$$
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$$\rho \sim \text{HalfNormal}(3,0.25)$$

$$G \sim \text{MVNormal}(\nu, \mathbf{K}_G)$$
$$\nu_i = \alpha_G + \beta_{MG} M_i$$
$$\mathbf{K}_G = \eta_G^2 \exp(-\rho_G d_{i,j})$$
$$\alpha_G \sim \text{Normal}(0,1)$$
$$\beta_{MG} \sim \text{Normal}(0,0.5)$$
$$\eta_G^2 \sim \text{HalfNormal}(1,0.25)$$
$$\rho_G \sim \text{HalfNormal}(3,0.25)$$

$$M \sim \text{MVNormal}(0,\mathbf{K}_M)$$
$$\mathbf{K}_M = \eta_M^2 \exp(-\rho_M d_{i,j})$$
$$\eta_M^2 \sim \text{HalfNormal}(1,0.25)$$
$$\rho_M \sim \text{HalfNormal}(3,0.25)$$

(2) Impute G and M ignoring models for each

$$\begin{tikzcd} G \arrow[rr] & & B \\ & M \arrow[ur] & \\ & h \arrow[r] & u \arrow[uu, bend right] \end{tikzcd}$$

$$G \qquad u \leftarrow h \qquad M$$

$$u \leftarrow h \qquad M$$

$$B \sim \text{MVNormal}(\mu, \mathbf{K})$$

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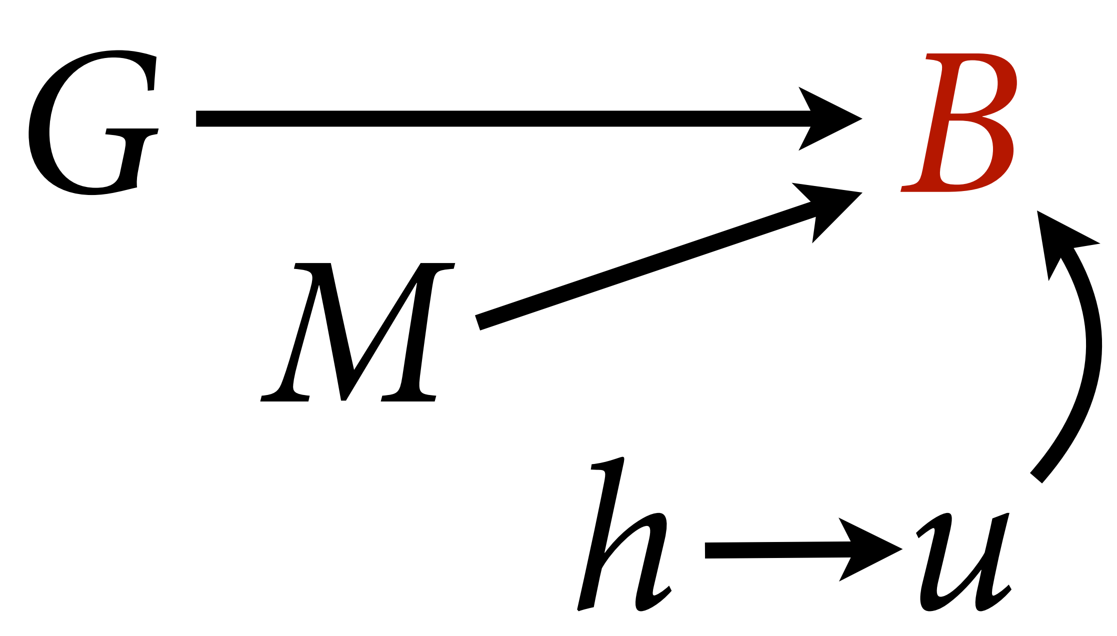
$$\eta^2 \sim \text{HalfNormal}(1,0.25)$$

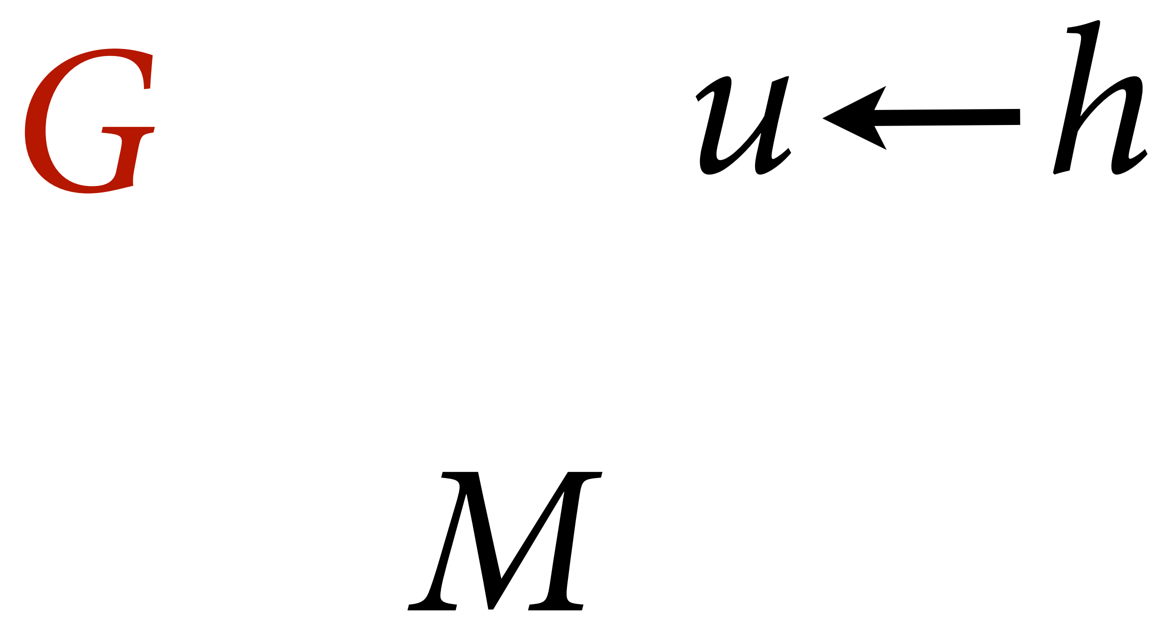
$$\rho \sim \text{HalfNormal}(3,0.25)$$

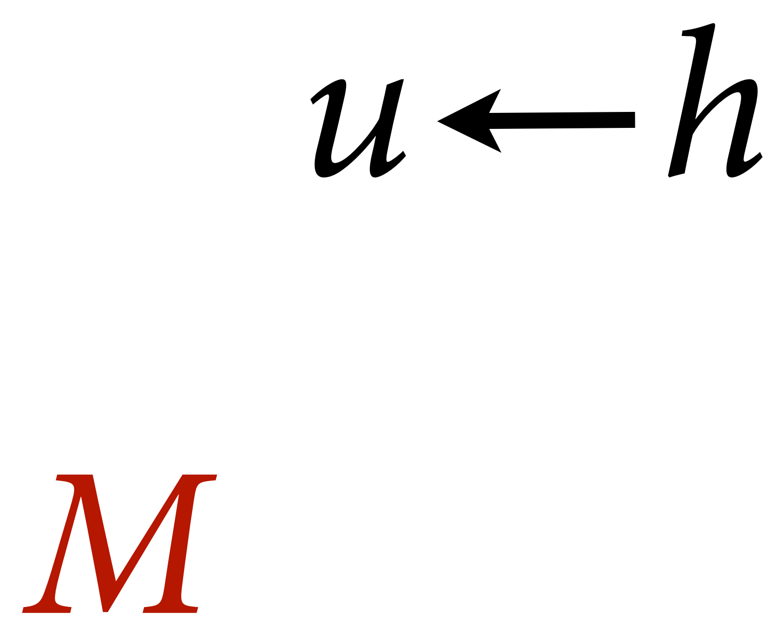
$$G_i \sim \text{Normal}(0,1)$$

$$M_i \sim \text{Normal}(0,1)$$

(2) Impute G and M ignoring models for each







$$B \sim \text{MVNormal}(\mu, \mathbf{K})$$
$$\mu_i = \alpha + \beta_G G_i + \beta_M M_i$$
$$\mathbf{K} = \eta^2 \exp(-\rho d_{i,j})$$
$$\alpha \sim \text{Normal}(0,1)$$
$$\beta_G, \beta_M \sim \text{Normal}(0,0.5)$$
$$\eta^2 \sim \text{HalfNormal}(1,0.25)$$
$$\rho \sim \text{HalfNormal}(3,0.25)$$

$$G_i \sim \text{Normal}(0,1)$$

$$M_i \sim \text{Normal}(0,1)$$

When G_i observed, likelihood for standardized variable

When G_i missing, prior

(2) Impute G and M ignoring models for each

```
# imputation ignoring models of M and G
fMBG_OU <- alist(
    B ~ multi_normal( mu , K ),
    mu <- a + bM*M + bG*G,
    matrix[N_spp,N_spp]:K <- cov_GPL1(Dmat,etasq,rho,0.01),
    M ~ normal(0,1),
    G ~ normal(0,1),
    a ~ normal( 0 , 1 ),
    c(bM,bG) ~ normal( 0 , 0.5 ),
    etasq ~ half_normal(1,0.25),
    rho ~ half_normal(3,0.25)
)
mBMG_OU <- ulam( fMBG_OU , data=dat_all,chains=4,cores=4 )
```

$$B \sim \text{MVNormal}(\mu, \mathbf{K})$$

$$\mu_i = \alpha + \beta_G G_i + \beta_M M_i$$

$$\mathbf{K} = \eta^2 \exp(-\rho d_{i,j})$$

$$G_i \sim \text{Normal}(0,1)$$

$$M_i \sim \text{Normal}(0,1)$$

$$\alpha \sim \text{Normal}(0,1)$$

$$\beta_G, \beta_M \sim \text{Normal}(0,0.5)$$

$$\eta^2 \sim \text{HalfNormal}(1,0.25)$$

$$\rho \sim \text{HalfNormal}(3,0.25)$$

(2) Impute G and M ignoring models for each

```
# imputation ignoring models of M and G
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    M ~ normal(0,1),
    G ~ normal(0,1),
    a ~ normal( 0 , 1 ),
    c(bM,bG) ~ normal( 0 , 0.5 ),
    etasq ~ half_normal(1,0.25),
    rho ~ half_normal(3,0.25)
)
mBMG_OU <- ulam( fMBG_OU , data=dat_all,chains=4,cores=4 )
```
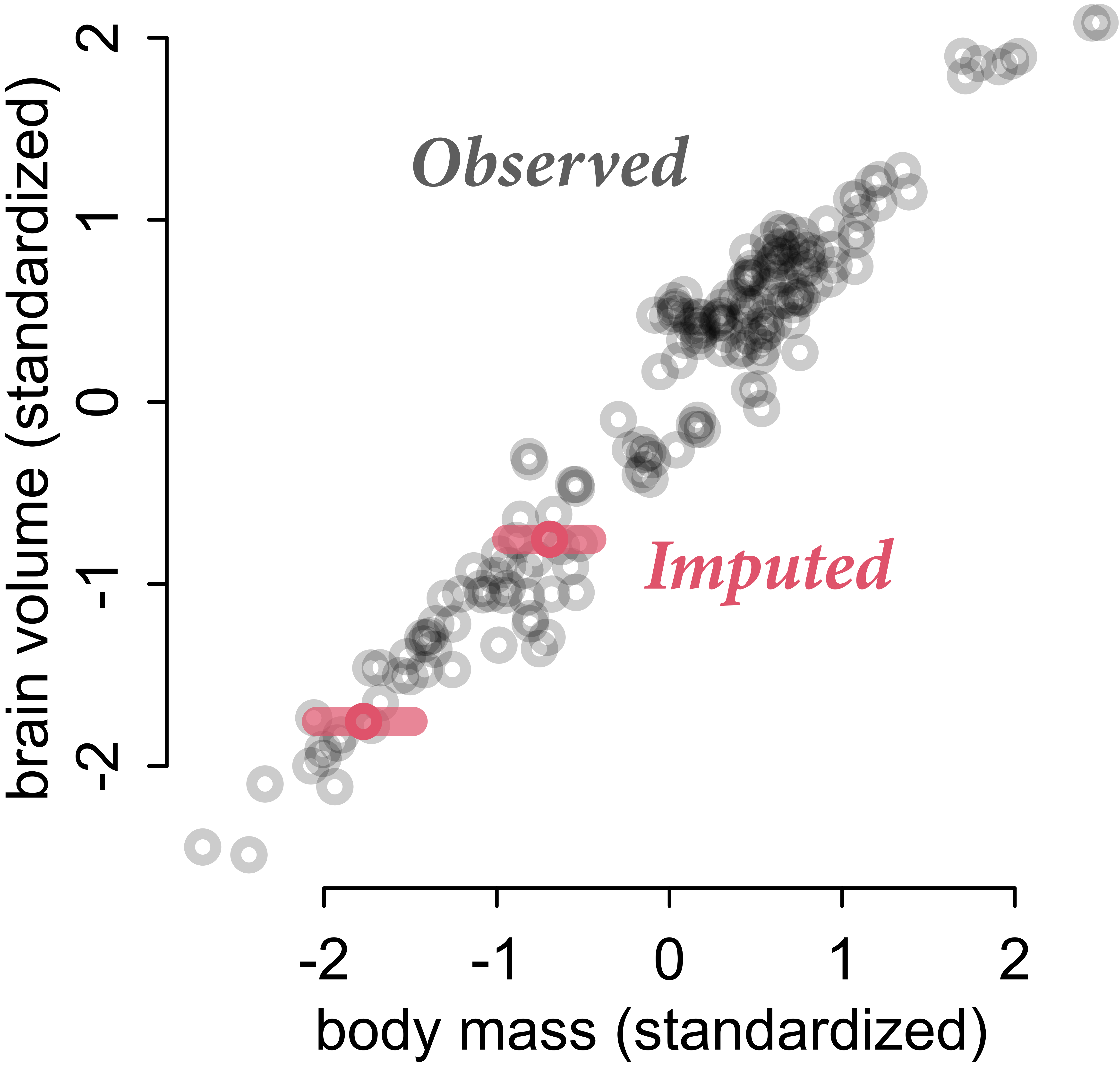

```
> precis(mBMG_OU,2)
               mean   sd  5.5% 94.5% n_eff Rhat4
a             -0.10 0.08 -0.22  0.02  4569     1
bG             0.01 0.02 -0.01  0.04  2816     1
bM             0.82 0.03  0.78  0.86  3949     1
etasq          0.04 0.01  0.03  0.05  3733     1
rho            2.75 0.26  2.36  3.17  3931     1
M_impute[1]   -1.77 0.18 -2.04 -1.48  5087     1
M_impute[2]   -0.69 0.15 -0.94 -0.45  5065     1
G_impute[1]    0.01 1.01 -1.61  1.63  6311     1
G_impute[2]    0.05 1.02 -1.55  1.67  5616     1
G_impute[3]    0.03 1.04 -1.64  1.71  4876     1
G_impute[4]    0.05 0.95 -1.45  1.56  4978     1
G_impute[5]    0.09 1.03 -1.53  1.69  5244     1
G_impute[6]    0.18 1.02 -1.48  1.79  5414     1
G_impute[7]    0.06 1.00 -1.57  1.65  6207     1
G_impute[8]    0.13 1.00 -1.47  1.72  6329     1
G_impute[9]    0.01 0.98 -1.62  1.62  5942     1
G_impute[10]   0.03 1.00 -1.56  1.67  5197     1
G_impute[11]   0.06 0.95 -1.47  1.60  5057     1
G_impute[12]   0.08 0.97 -1.49  1.63  4907     1
G_impute[13]  -0.03 1.01 -1.65  1.58  5958     1
G_impute[14]   0.04 0.99 -1.52  1.65  5538     1
G_impute[15]  -0.18 1.00 -1.77  1.37  5705     1
G_impute[16]  -0.09 0.97 -1.63  1.45  4026     1
G_impute[17]  -0.04 1.02 -1.74  1.58  6465     1
G_impute[18]  -0.09 1.03 -1.73  1.54  5632     1
G_impute[19]  -0.19 1.01 -1.78  1.45  4251     1
G_impute[20]  -0.01 1.02 -1.62  1.65  6602     1
G_impute[21]  -0.07 0.95 -1.56  1.47  5640     1
G_impute[22]  -0.04 0.99 -1.60  1.55  6602     1
G_impute[23]   0.14 1.03 -1.45  1.82  5644     1
G_impute[24]  -0.07 0.99 -1.65  1.46  4249     1
G_impute[25]  -0.06 1.04 -1.72  1.62  5993     1
G_impute[26]   0.19 1.01 -1.40  1.78  4973     1
G_impute[27]   0.08 1.04 -1.54  1.70  5765     1
G_impute[28]  -0.07 0.99 -1.66  1.52  3609     1
G_impute[29]   0.04 0.99 -1.55  1.63  4542     1
G_impute[30]  -0.01 0.99 -1.61  1.56  4821     1
G_impute[31]   0.01 1.02 -1.65  1.66  6602     1
G_impute[32]   0.08 1.01 -1.51  1.66  5420     1
G_impute[33]  -0.02 0.99 -1.58  1.60  4241     1
```

5)

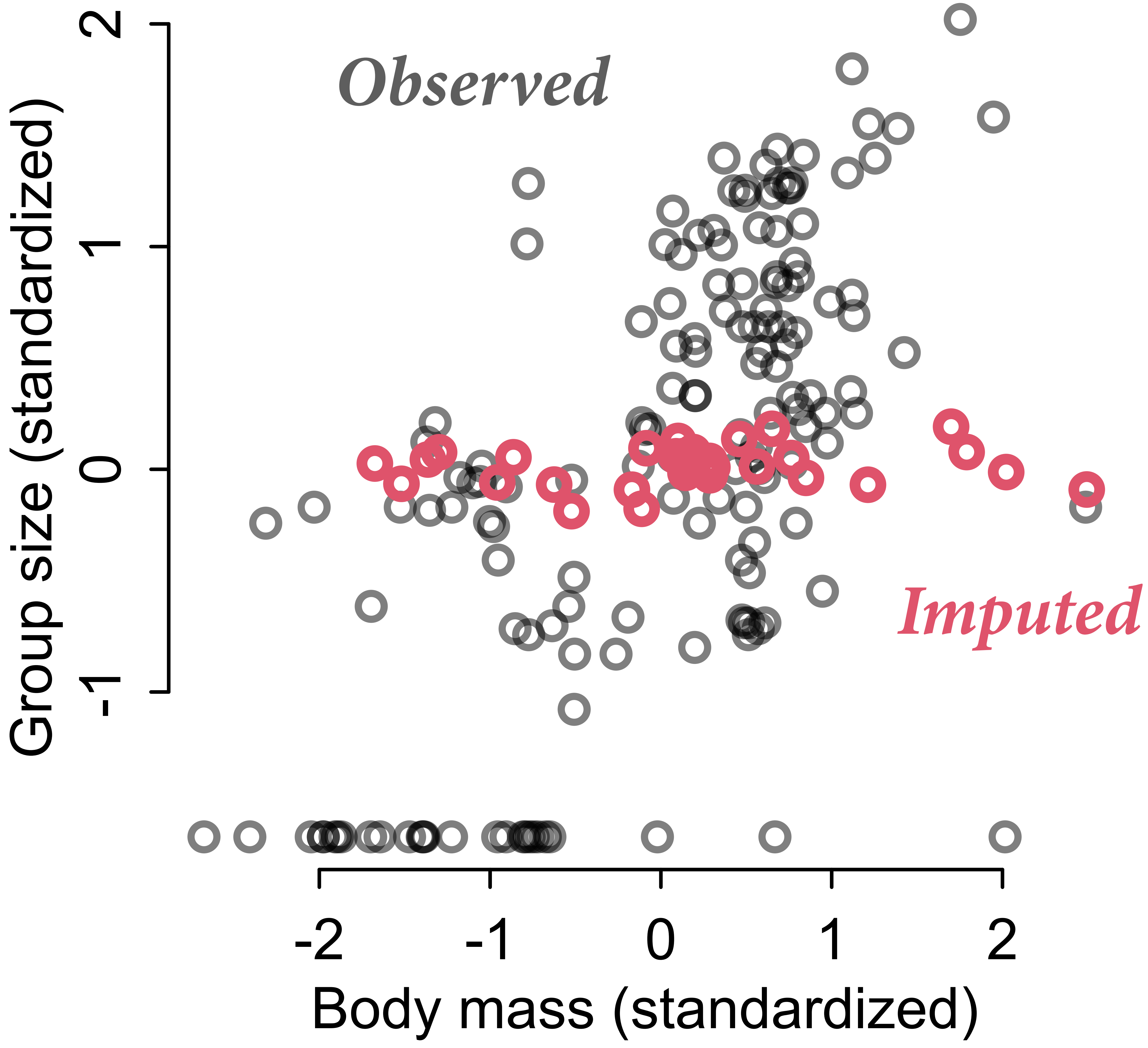
5)

(2) Impute G and M ignoring models for each



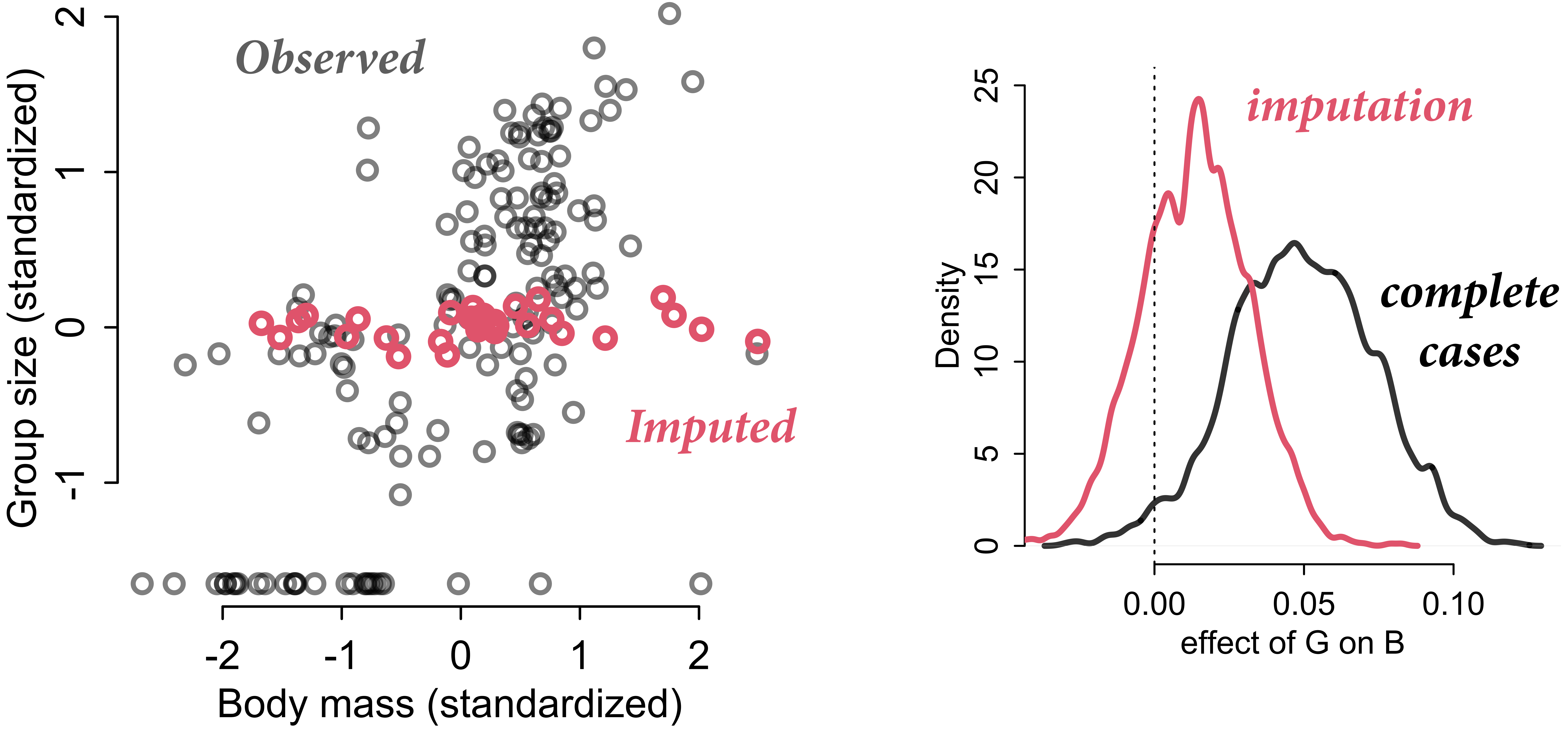
Because M strongly associated with B, **imputed M values** follow the regression relationship

(2) Impute G and M ignoring models for each

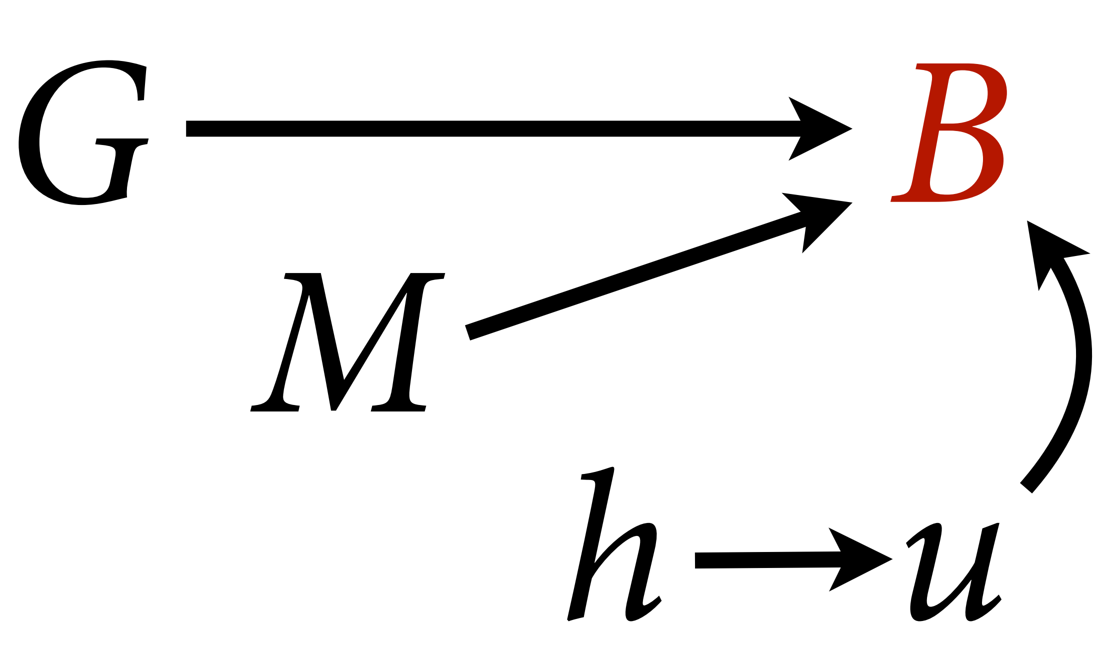


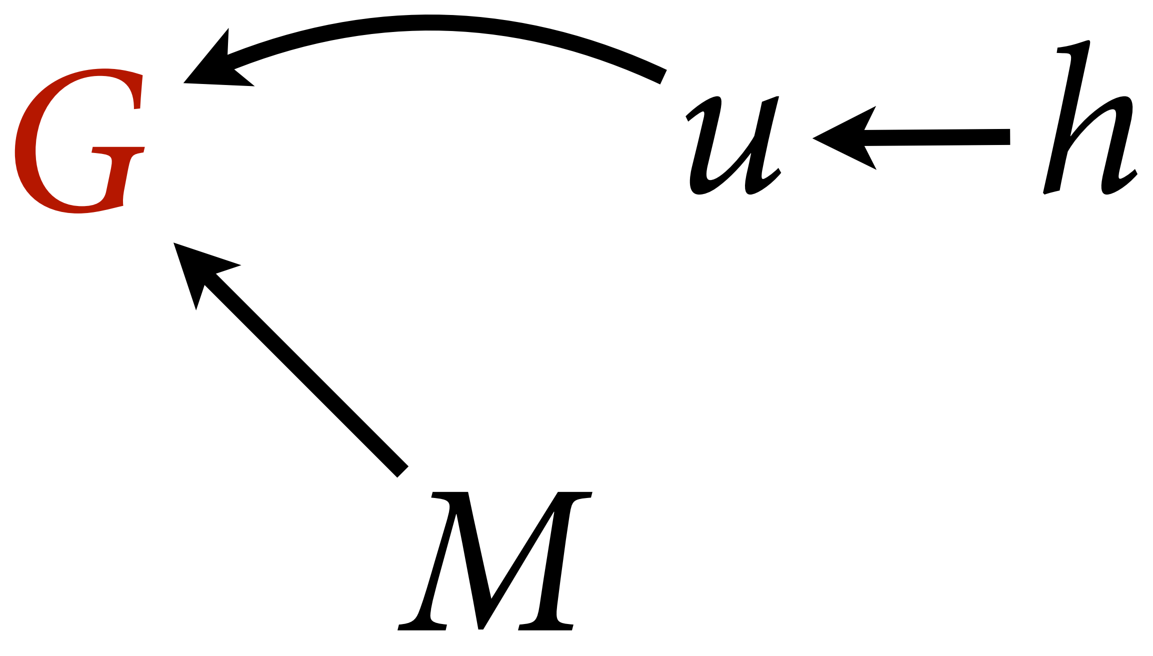
Because association between M and G not modeled, **imputed G values** do not follow the regression relationship

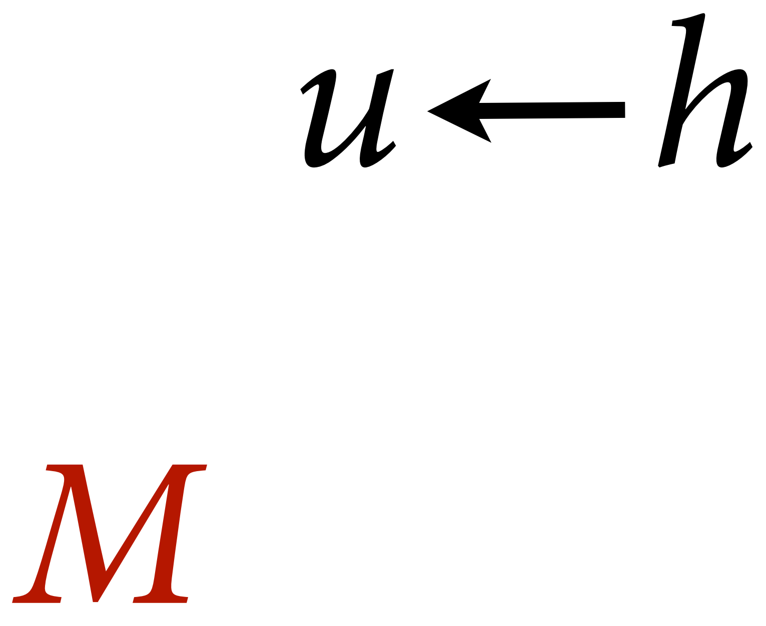
(2) Impute *G* and *M* ignoring models for each



(3) Impute G using model







$$
\begin{aligned}
B &\sim \mathrm{MVNormal}(\mu, \mathbf{K}) \\
\mu_i &= \alpha + \beta_G G_i + \beta_M M_i \\
\mathbf{K} &= \eta^2 \exp(-\rho d_{i,j}) \\
\alpha &\sim \mathrm{Normal}(0,1) \\
\beta_G, \beta_M &\sim \mathrm{Normal}(0,0.5) \\
\eta^2 &\sim \mathrm{HalfNormal}(1,0.25) \\
\rho &\sim \mathrm{HalfNormal}(3,0.25)
\end{aligned}
$$

$$
\begin{aligned}
G &\sim \mathrm{MVNormal}(\nu, \mathbf{K}_G) \\
\nu_i &= \alpha_G + \beta_{MG} M_i \\
\mathbf{K}_G &= \eta_G^2 \exp(-\rho_G d_{i,j}) \\
\alpha_G &\sim \mathrm{Normal}(0,1) \\
\beta_{MG} &\sim \mathrm{Normal}(0,0.5) \\
\eta_G^2 &\sim \mathrm{HalfNormal}(1,0.25) \\
\rho_G &\sim \mathrm{HalfNormal}(3,0.25)
\end{aligned}
$$

$$M_i \sim \mathrm{Normal}(0,1)$$

(3) Impute G using model

```
# no phylogeny on G but have submodel M -> G
mBMG_OU_G <- ulam(
    alist(
        B ~ multi_normal( mu , K ),
        mu <- a + bM*M + bG*G,
        G ~ normal(nu,sigma),
        nu <- aG + bMG*M,
        M ~ normal(0,1),
        matrix[N_spp,N_spp]:K <- cov_GPL1(Dmat,etasq,rho,0.01),
        c(a,aG) ~ normal( 0 , 1 ),
        c(bM,bG,bMG) ~ normal( 0 , 0.5 ),
        c(etasq) ~ half_normal(1,0.25),
        c(rho) ~ half_normal(3,0.25),
        sigma ~ exponential(1)
    ), data=dat_all , chains=4 , cores=4 , sample=TRUE )

# phylogeny information for G imputation (but no M -> G model)
mBMG_OU2 <- ulam(
    alist(
        B ~ multi_normal( mu , K ),
        mu <- a + bM*M + bG*G,
        M ~ normal(0,1),
        G ~ multi_normal( 'rep_vector(0,N_spp)' ,KG),
        matrix[N_spp,N_spp]:K <- cov_GPL1(Dmat,etasq,rho,0.01),
        matrix[N_spp,N_spp]:KG <- cov_GPL1(Dmat,etasqG,rhoG,0.01),
        a ~ normal( 0 , 1 ),
        c(bM,bG) ~ normal( 0 , 0.5 ),
        c(etasq,etasqG) ~ half_normal(1,0.25),
        c(rho,rhoG) ~ half_normal(3,0.25)
    ), data=dat_all , chains=4 , cores=4 , sample=TRUE )
```

Just M -> G model

$$G \sim \text{MVNormal}(\nu, \mathbf{I}\sigma^2)$$

$$\nu_i = \alpha_G + \beta_{MG} M_i$$

Just phylogeny

$$G \sim \text{MVNormal}(0, \mathbf{K}_G)$$

$$\mathbf{K}_G = \eta_G^2 \exp(-\rho_G d_{i,j})$$

```r
# no phylogeny on G but have submodel M -> G
mBMG_OU_G <- ulam(
    alist(
        B ~ multi_normal( mu , K ),
        mu <- a + bM*M + bG*G,
        G ~ normal(nu,sigma),
        nu <- aG + bMG*M,
        M ~ normal(0,1),
        matrix[N_spp,N_spp]:K <- cov_GPL1(Dmat,etasq,rho,0.01),
        c(a,aG) ~ normal( 0 , 1 ),
        c(bM,bG,bMG) ~ normal( 0 , 0.5 ),
        c(etasq) ~ half_normal(1,0.25),
        c(rho) ~ half_normal(3,0.25),
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    ), data=dat_all , chains=4 , cores=4 , sample=TRUE )

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    alist(
        B ~ multi_normal( mu , K ),
        mu <- a + bM*M + bG*G,
        M ~ normal(0,1),
        G ~ multi_normal( 'rep_vector(0,N_spp)' ,KG),
        matrix[N_spp,N_spp]:K <- cov_GPL1(Dmat,etasq,rho,0.01),
        matrix[N_spp,N_spp]:KG <- cov_GPL1(Dmat,etasqG,rhoG,0.01),
        a ~ normal( 0 , 1 ),
        c(bM,bG) ~ normal( 0 , 0.5 ),
        c(etasq,etasqG) ~ half_normal(1,0.25),
        c(rho,rhoG) ~ half_normal(3,0.25)
    ), data=dat_all , chains=4 , cores=4 , sample=TRUE )
```

(3) Impute G using model

Just M –> G model

$$G \sim \text{MVNormal}(\nu, \mathbf{I}\sigma^2)$$

$$\nu_i = \alpha_G + \beta_{MG} M_i$$

Just phylogeny

$$G \sim \text{MVNormal}(0, \mathbf{K}_G)$$

$$\mathbf{K}_G = \eta_G^2 \exp(-\rho_G d_{i,j})$$

```
# no phylogeny on G but have submodel M -> G
mBMG_OU_G <- ulam(
    alist(
        B ~ multi_normal( mu , K ),
        mu <- a + bM*M + bG*G,
        G ~ normal(nu,sigma),
        nu <- aG + bMG*M,
        M ~ normal(0,1),
        matrix[N_spp,N_spp]:K <- cov_GPL1(Dmat,etasq,rho,0.01),
        c(a,aG) ~ normal( 0 , 1 ),
        c(bM,bG,bMG) ~ normal( 0 , 0.5 ),
        c(etasq) ~ half_normal(1,0.25),
        c(rho) ~ half_normal(3,0.25),
        sigma ~ exponential(1)
    ), data=dat_all , chains=4 , cores=4 , sample=TRUE )

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    alist(
        B ~ multi_normal( mu , K ),
        mu <- a + bM*M + bG*G,
        M ~ normal(0,1),
        G ~ multi_normal( 'rep_vector(0,N_spp)' ,KG),
        matrix[N_spp,N_spp]:K <- cov_GPL1(Dmat,etasq,rho,0.01),
        matrix[N_spp,N_spp]:KG <- cov_GPL1(Dmat,etasqG,rhoG,0.01),
        a ~ normal( 0 , 1 ),
        c(bM,bG) ~ normal( 0 , 0.5 ),
        c(etasq,etasqG) ~ half_normal(1,0.25),
        c(rho,rhoG) ~ half_normal(3,0.25)
    ), data=dat_all , chains=4 , cores=4 , sample=TRUE )
```

(3) Impute G using model

Just M -> G model

$$G \sim \text{MVNormal}(\nu, \mathbf{I}\sigma^2)$$

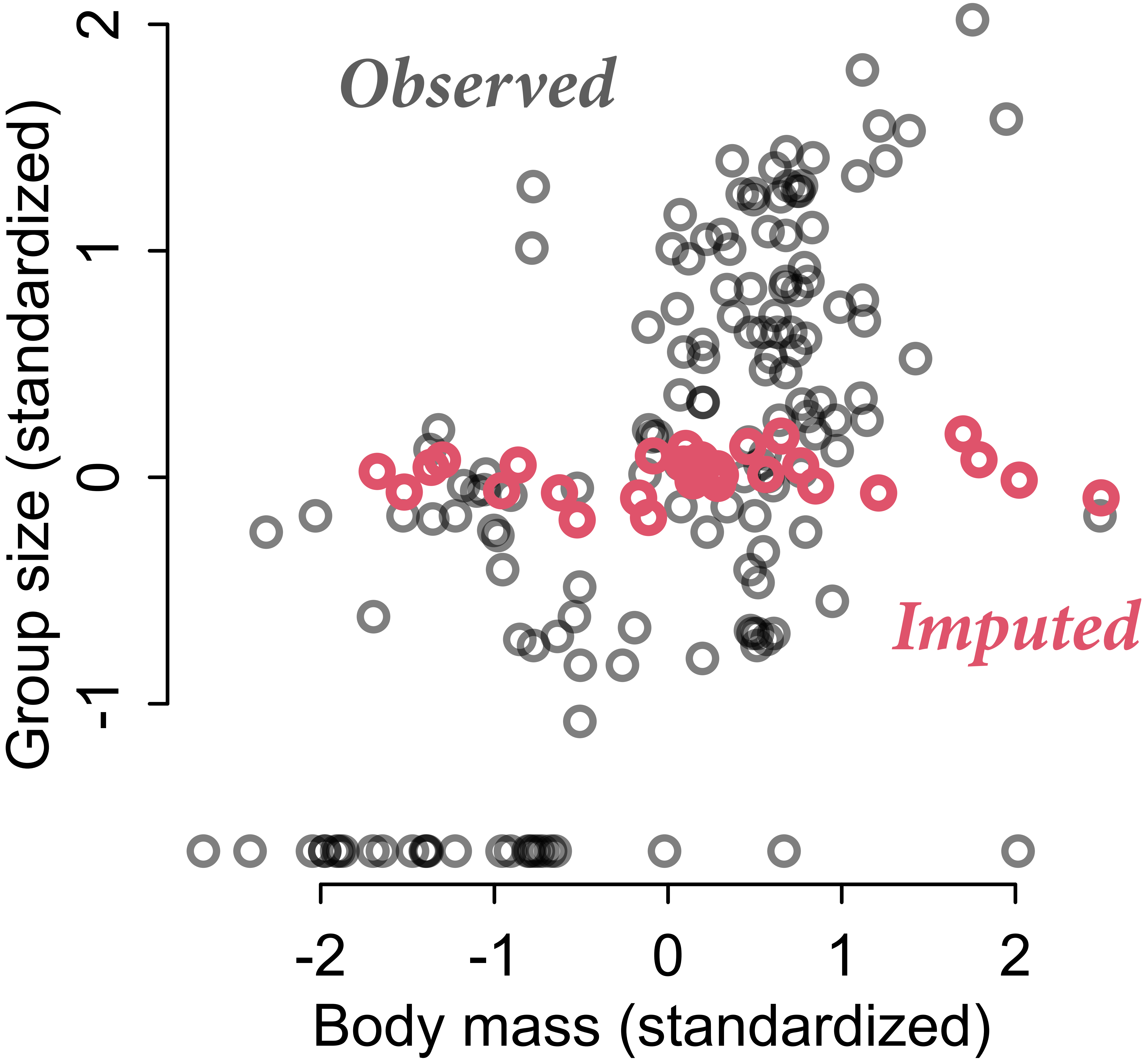
$$\nu_i = \alpha_G + \beta_{MG} M_i$$

Just phylogeny

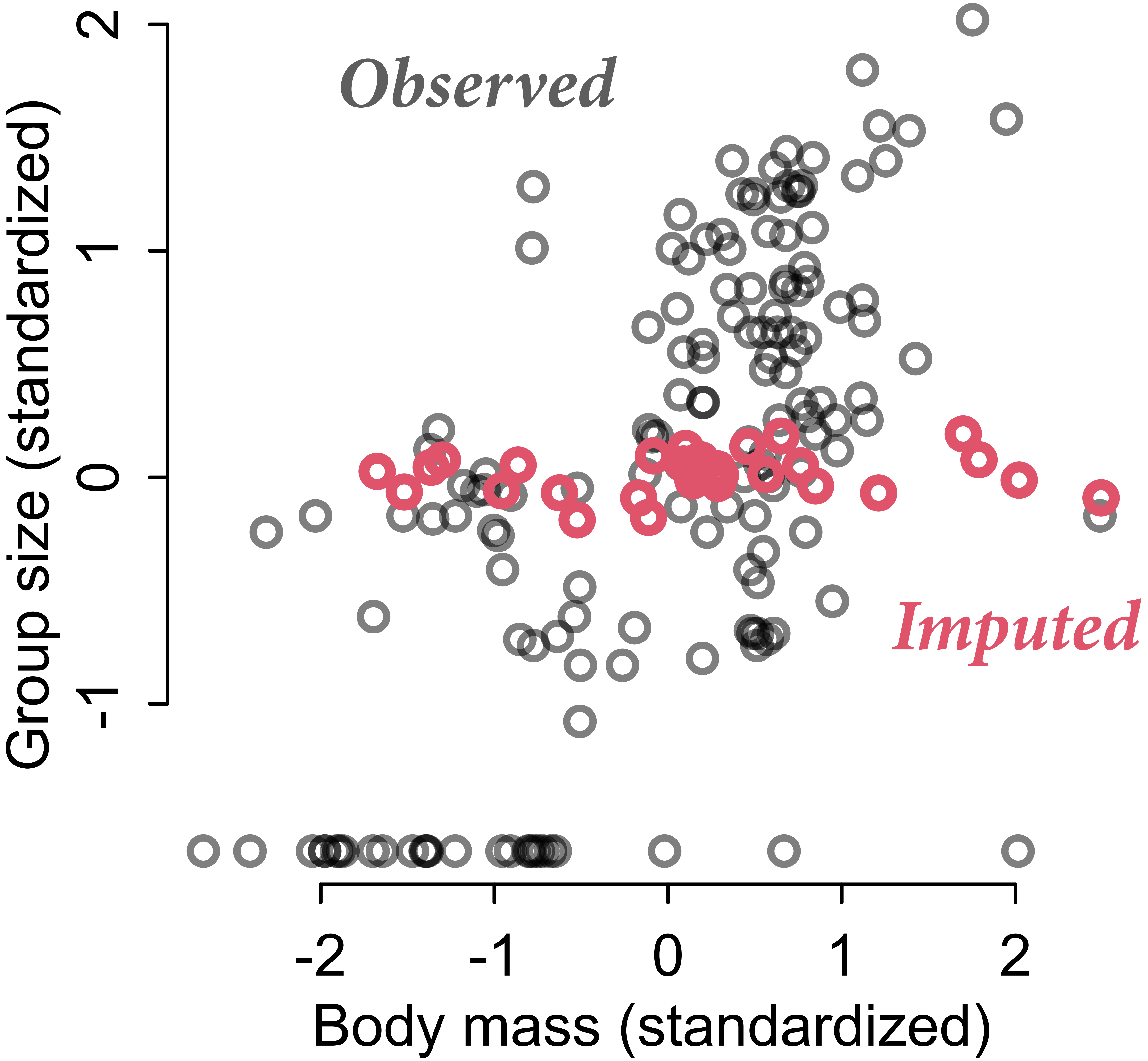
$$G \sim \text{MVNormal}(0, \mathbf{K}_G)$$

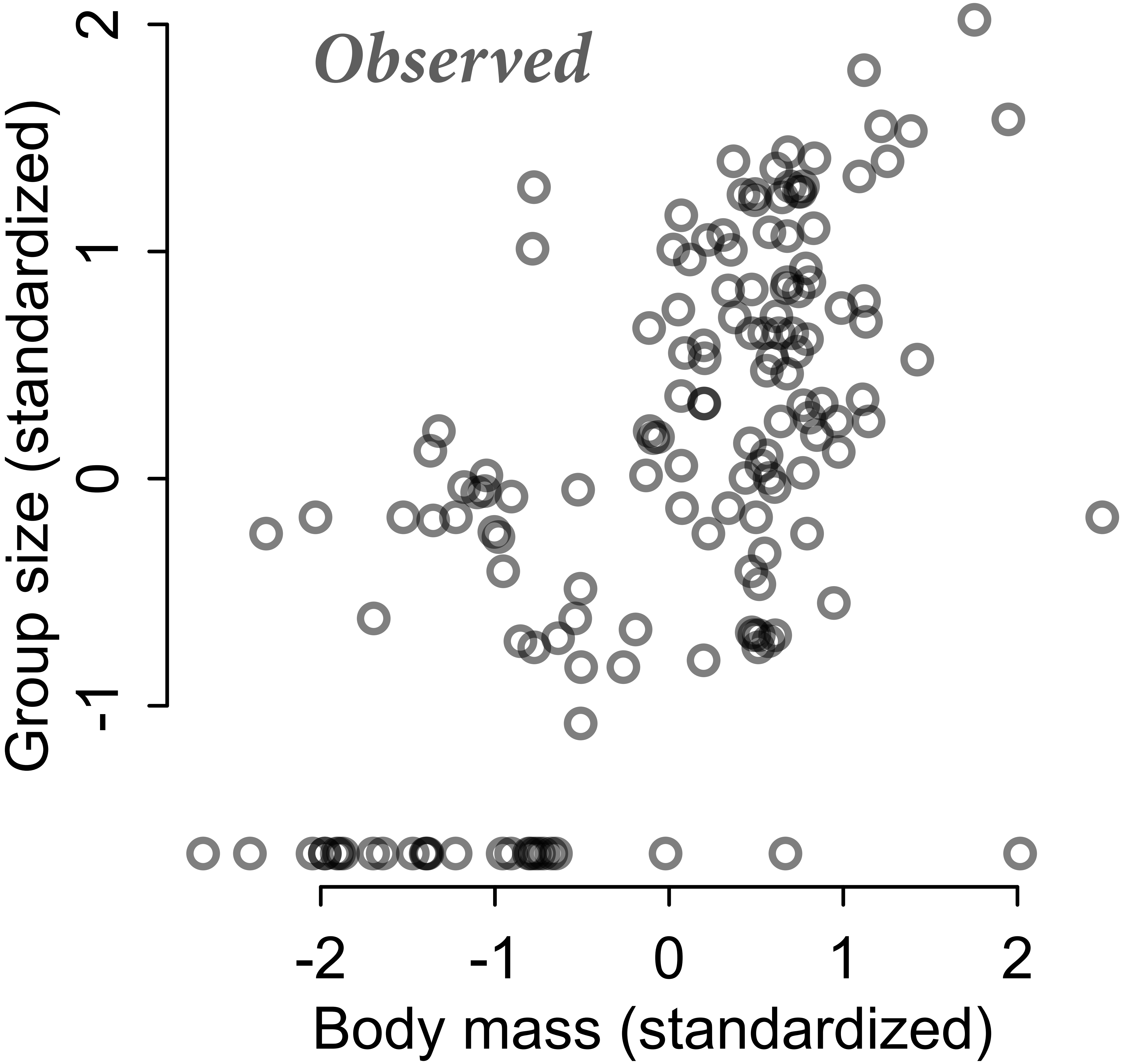
$$\mathbf{K}_G = \eta_G^2 \exp(-\rho_G d_{i,j})$$

(3) Impute G using model

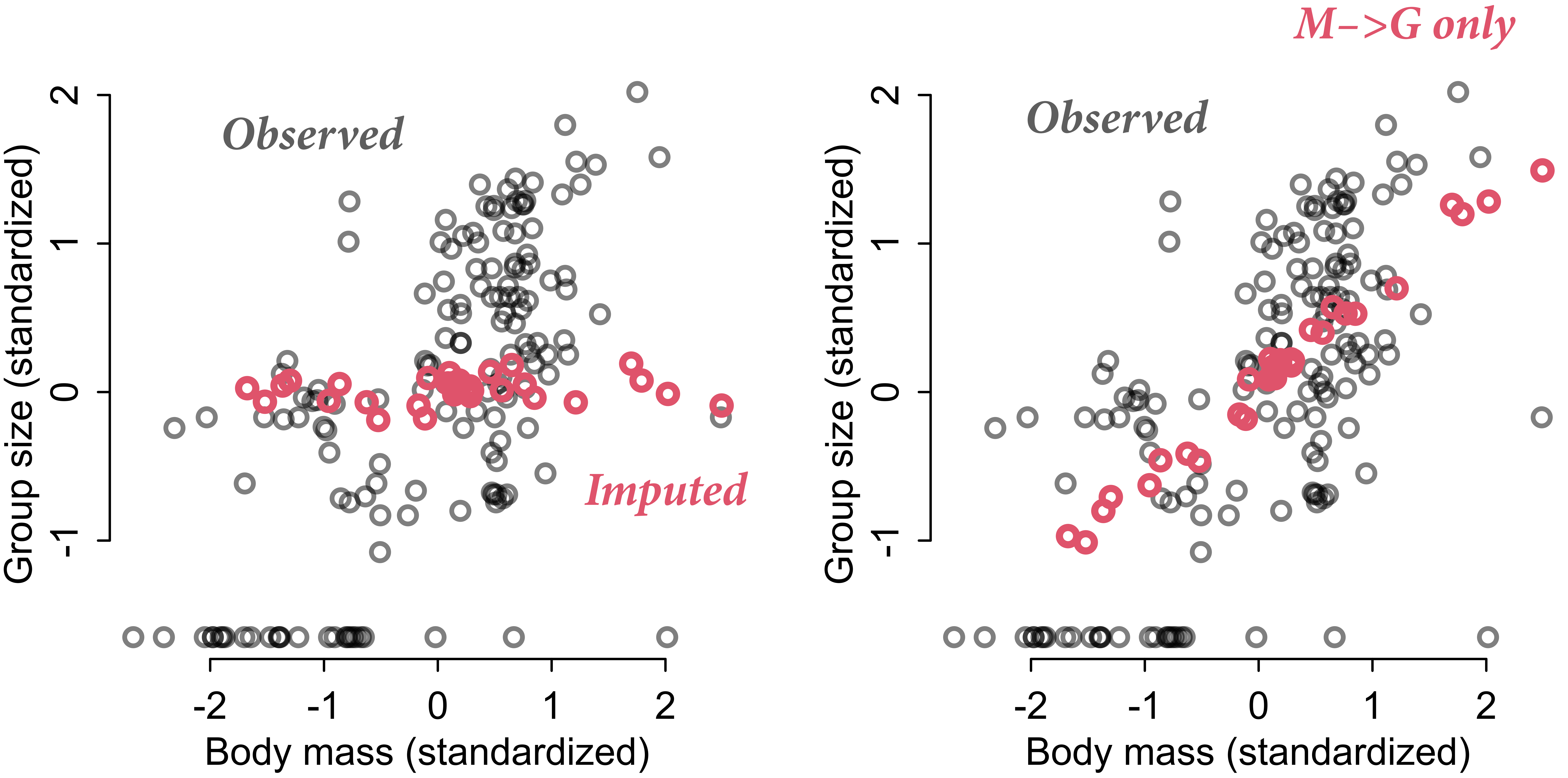


(3) Impute G using model

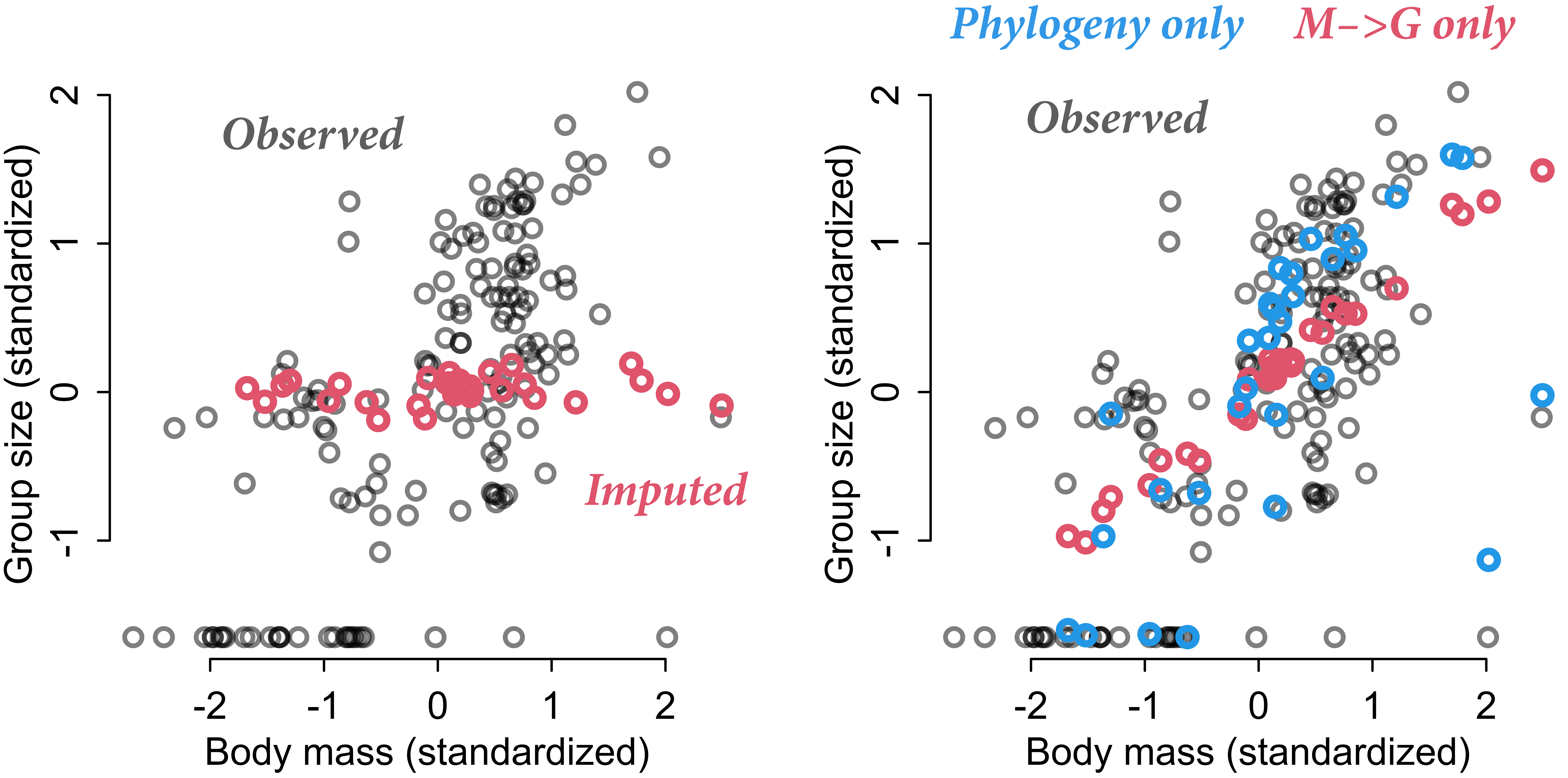




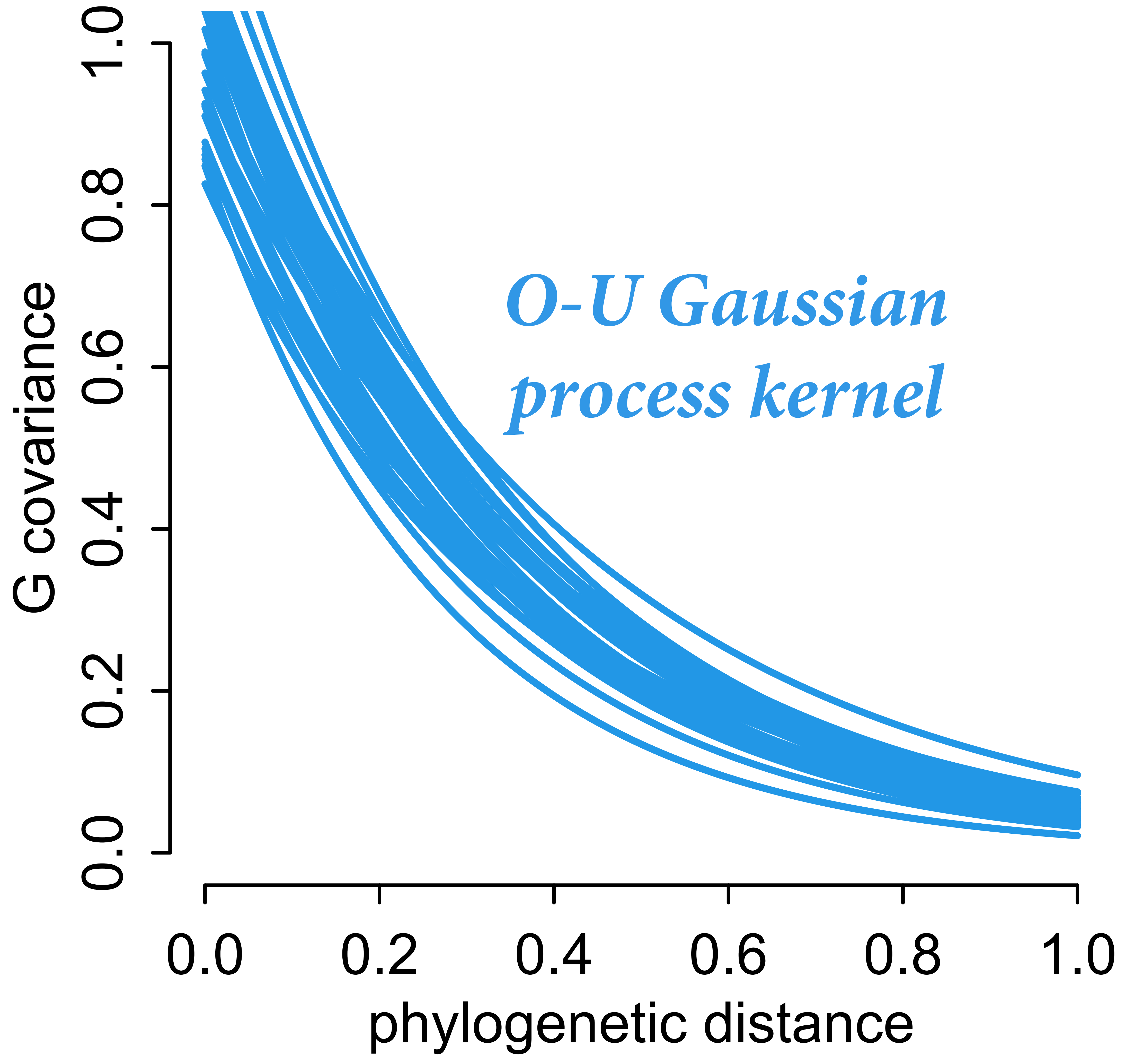
(3) Impute G using model

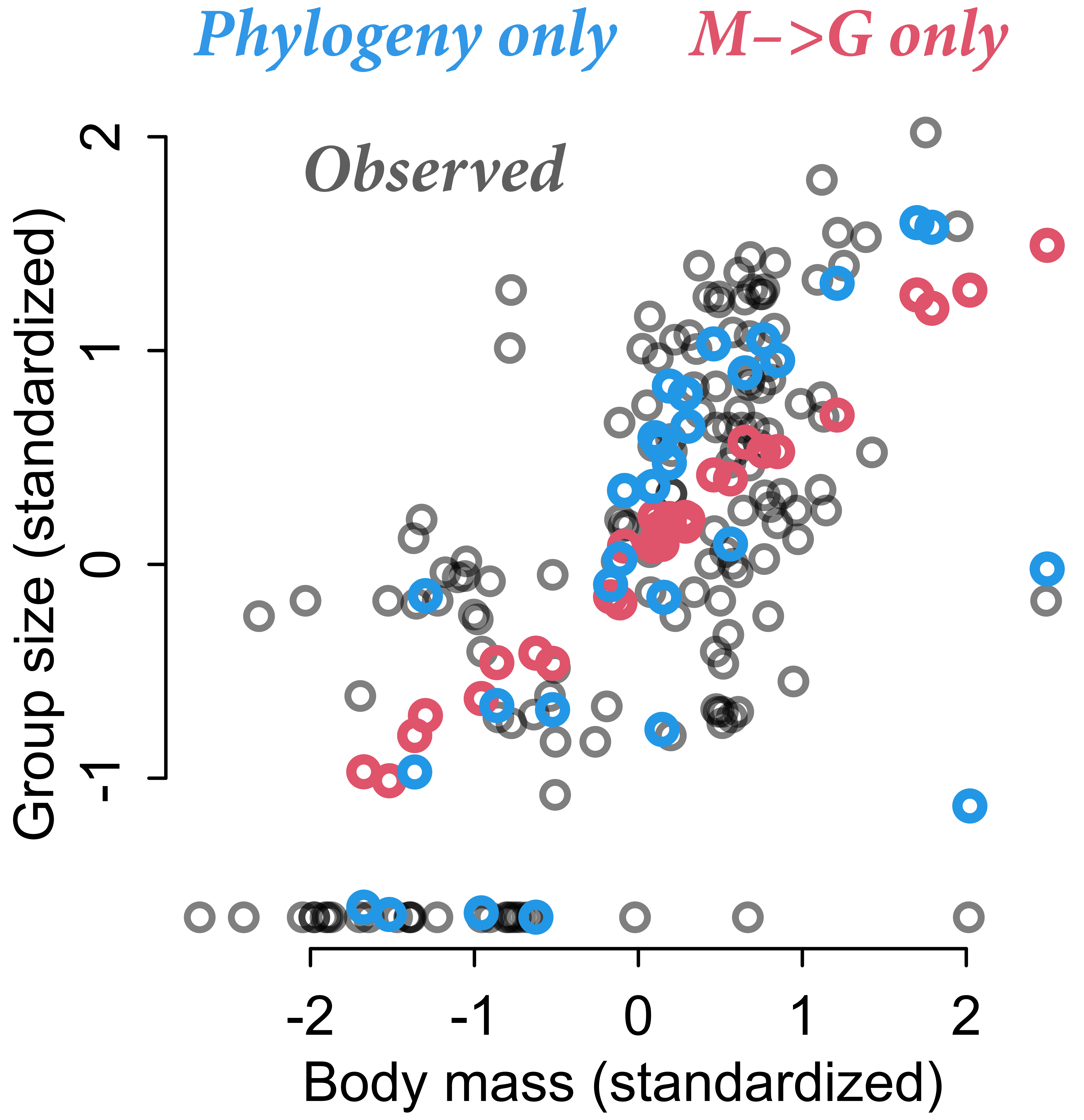


(3) Impute *G* using model

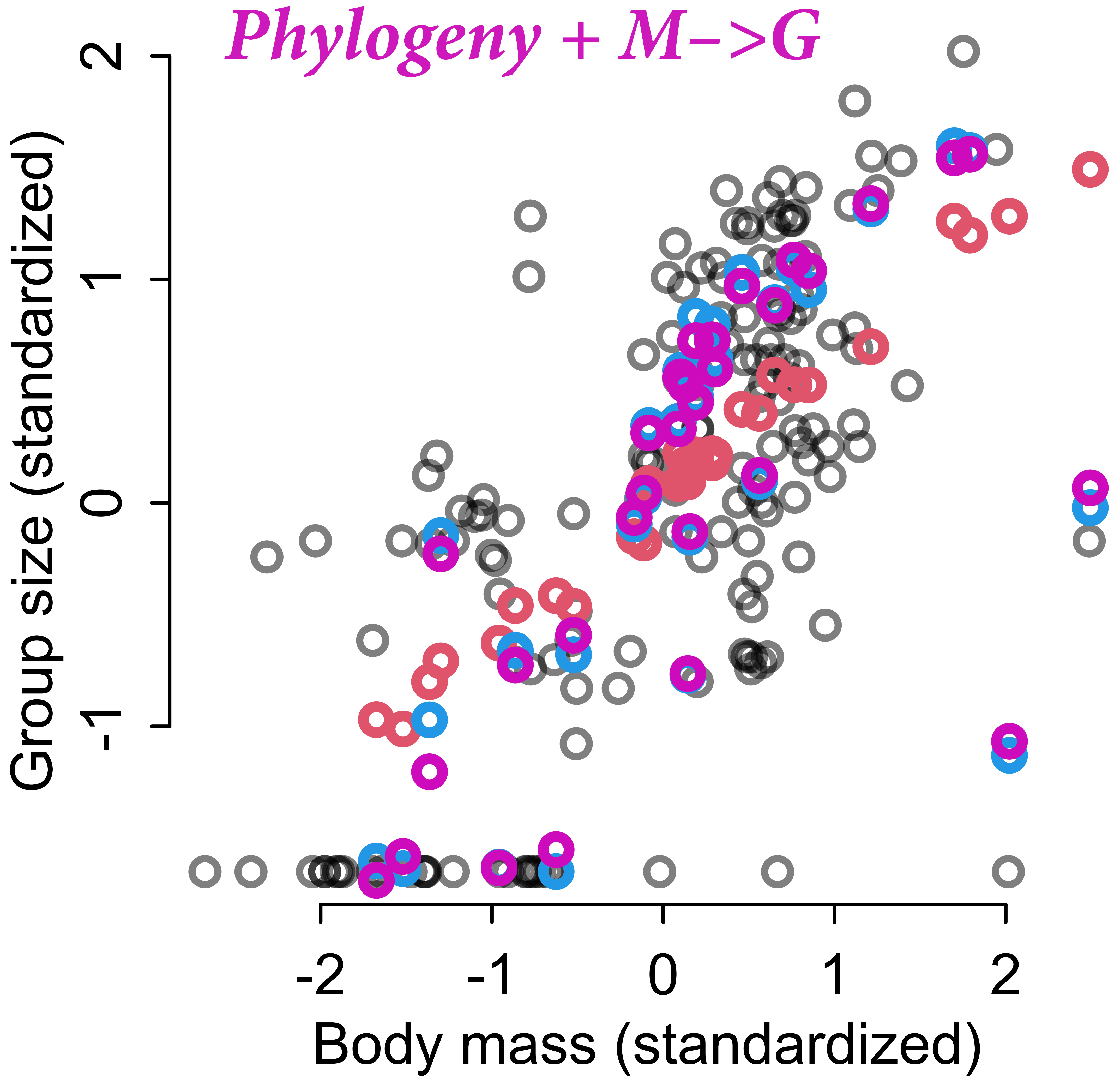


(3) Impute *G* using model

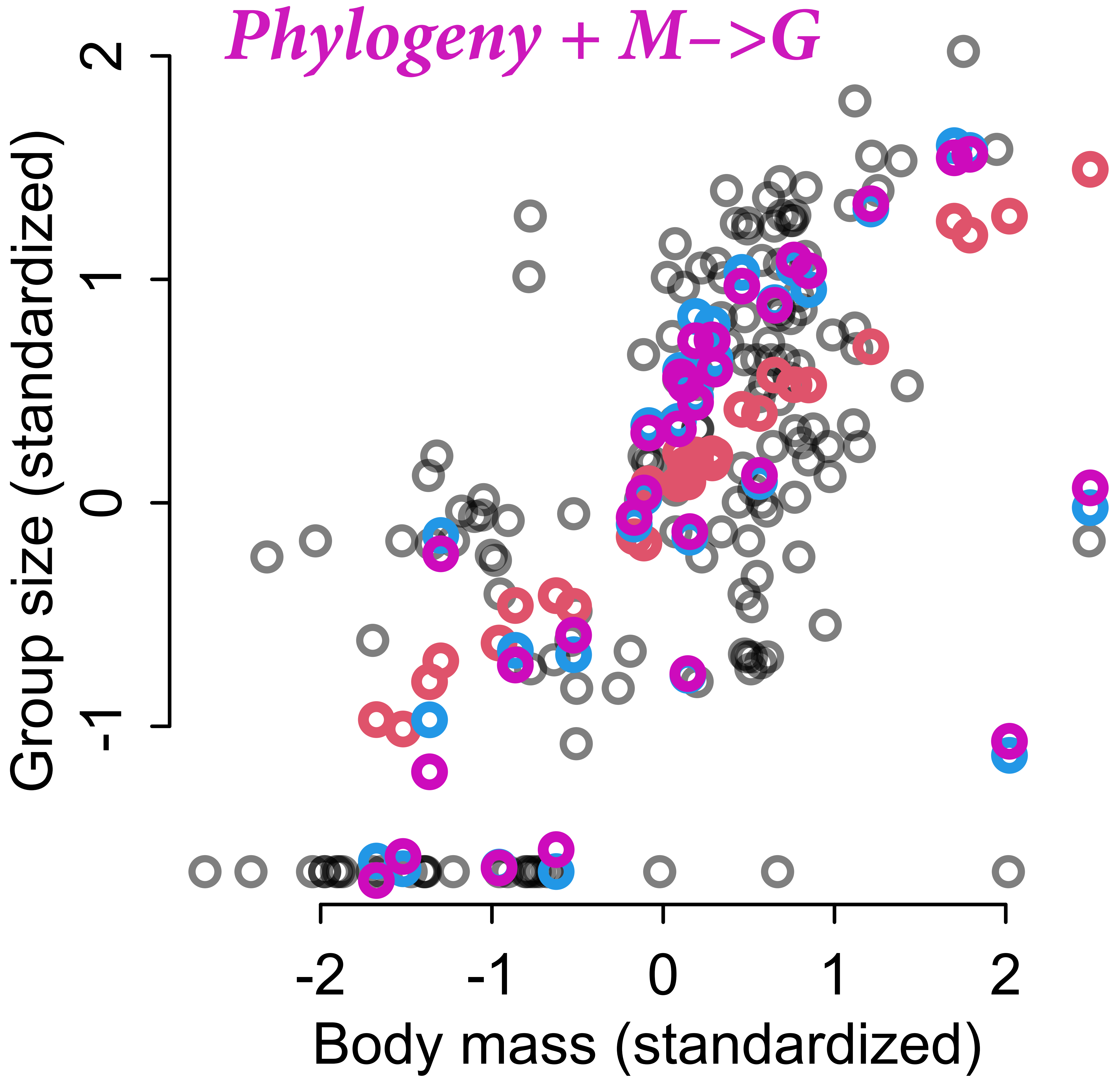


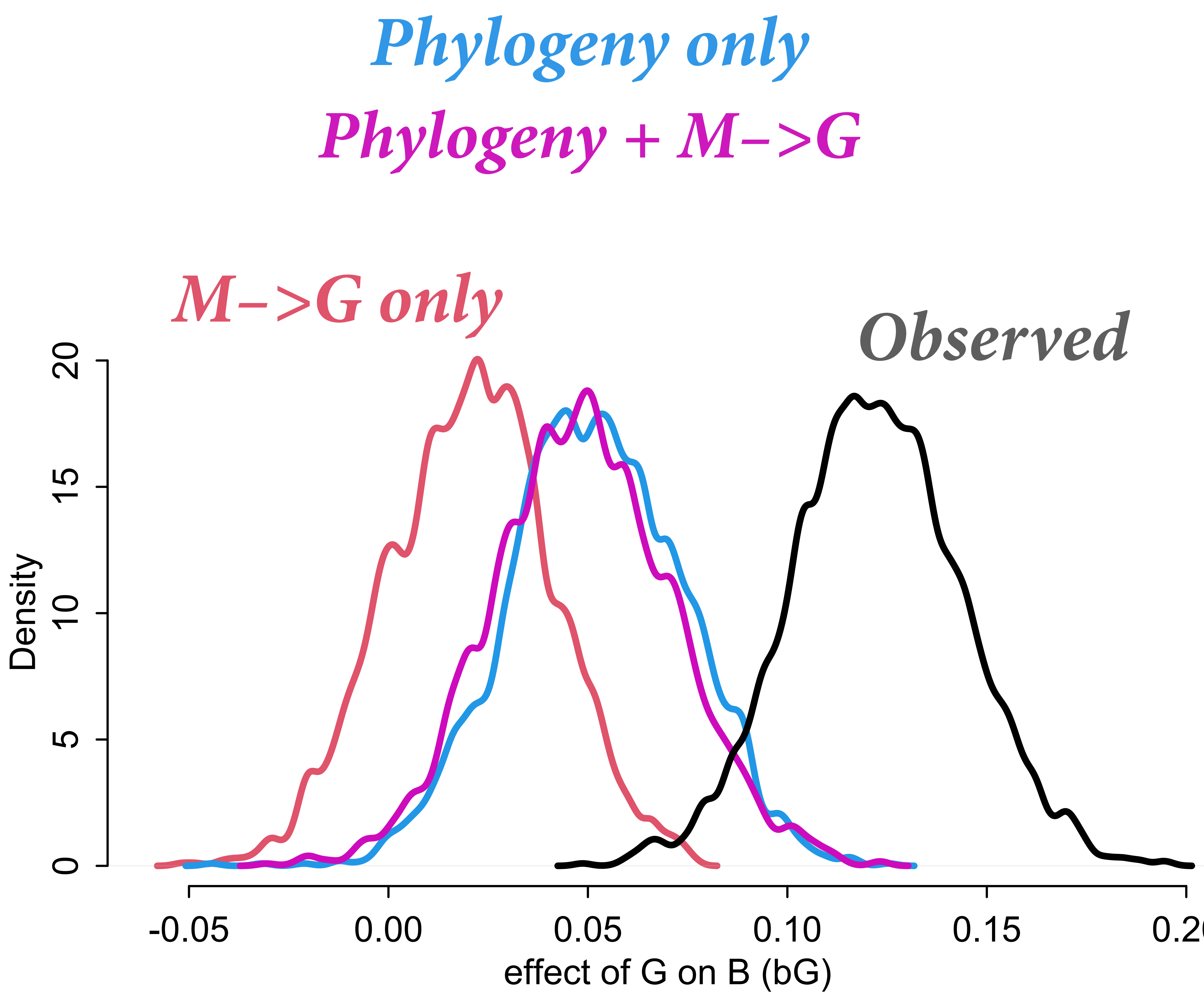


(3) Impute G using model



(3) Impute *G* using model





Draw the Missing Owl

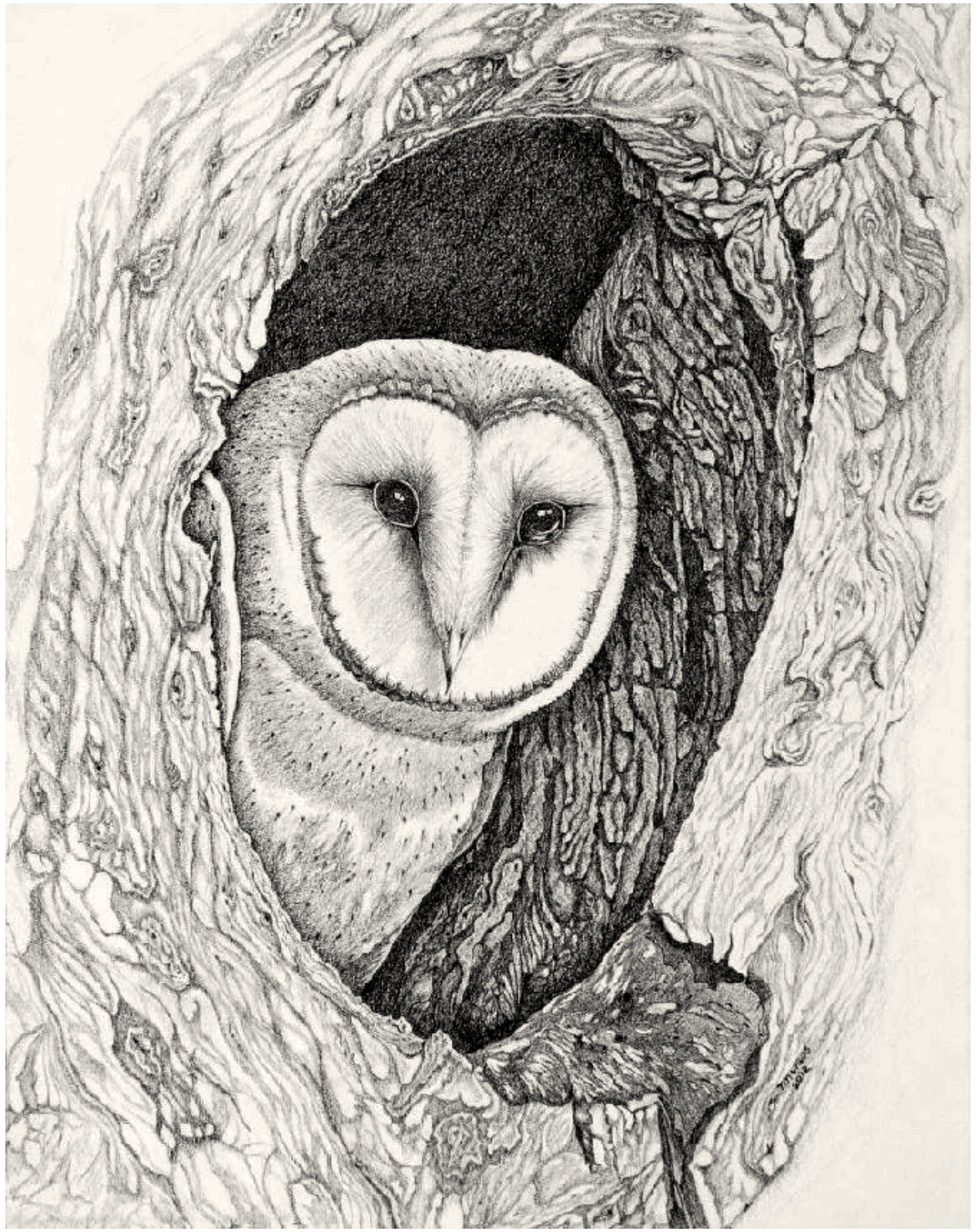
Let's take it slow…

(1) Ignore cases with missing B values (for now)

(2) Impute G and M ignoring models for each

(3) Impute G using model

(4) Impute B, G, M using model



(4) Impute B, G, M using model

$$\begin{tikzcd} G \arrow[rr] & & B \\ & M \arrow[ur] & \\ & h \arrow[r] & u \arrow[uu, bend right] \end{tikzcd}$$

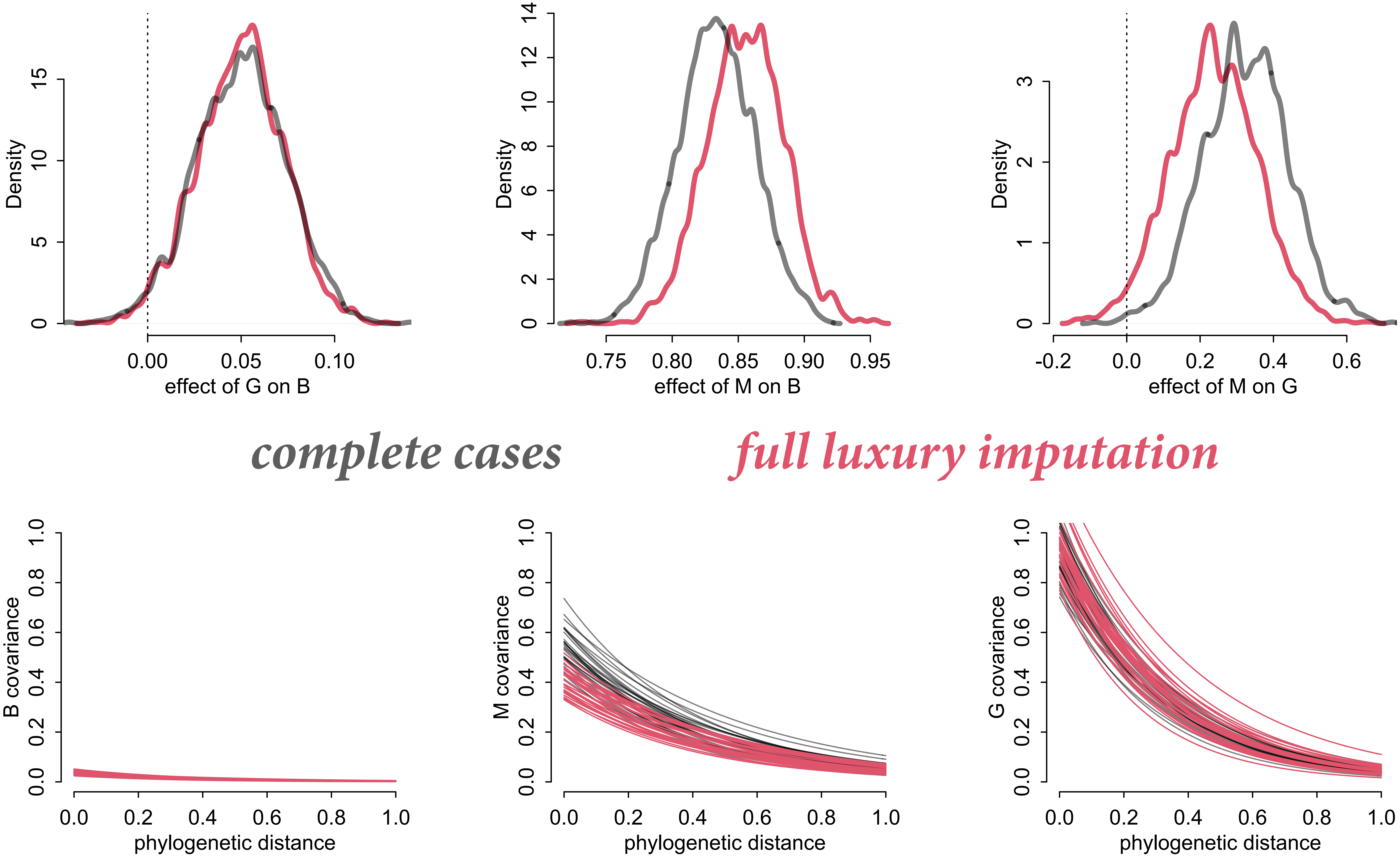
$$B \sim \text{MVNormal}(\mu, \mathbf{K})$$
$$\mu_i = \alpha + \beta_G G_i + \beta_M M_i$$
$$\mathbf{K} = \eta^2 \exp(-\rho d_{i,j})$$
$$\alpha \sim \text{Normal}(0,1)$$
$$\beta_G, \beta_M \sim \text{Normal}(0,0.5)$$
$$\eta^2 \sim \text{HalfNormal}(1,0.25)$$
$$\rho \sim \text{HalfNormal}(3,0.25)$$

$$\begin{tikzcd} G & & u \arrow[ll, bend right] & h \arrow[l] \\ & M \arrow[ul] & & \end{tikzcd}$$

$$G \sim \text{MVNormal}(\nu, \mathbf{K}_G)$$
$$\nu_i = \alpha_G + \beta_{MG} M_i$$
$$\mathbf{K}_G = \eta_G^2 \exp(-\rho_G d_{i,j})$$
$$\alpha_G \sim \text{Normal}(0,1)$$
$$\beta_{MG} \sim \text{Normal}(0,0.5)$$
$$\eta_G^2 \sim \text{HalfNormal}(1,0.25)$$
$$\rho_G \sim \text{HalfNormal}(3,0.25)$$

$$\begin{tikzcd} & u \arrow[dl, bend right] & h \arrow[l] \\ M & & \end{tikzcd}$$

$$M \sim \text{MVNormal}(0,\mathbf{K}_M)$$
$$\mathbf{K}_M = \eta_M^2 \exp(-\rho_M d_{i,j})$$
$$\eta_M^2 \sim \text{HalfNormal}(1,0.25)$$
$$\rho_M \sim \text{HalfNormal}(3,0.25)$$



Density
effect of G on B
0.00
0.05
0.10
0
5
10
15
Density
effect of M on B
0.75
0.80
0.85
0.90
0.95
0
2
4
6
8
10
12
14
Density
effect of M on G
-0.2
0.0
0.2
0.4
0.6
0
1
2
3
complete cases
full luxury imputation
B covariance
M covariance
G covariance
phylogenetic distance
0.0
0.2
0.4
0.6
0.8
1.0

Imputing Primates

Key idea: Missing values already have probability distributions

Think like a graph, not like a regression

Imputation without relationships among predictors risky

Even if doesn't change result, it's our duty